

Trends in Consciousness Research

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Abstract

This chapter offers an early-career researchers' perspective on the future of consciousness science across disciplines. Building on the foundational work surveyed earlier in this volume, we identify four leverage points where, in our eyes, the next decisive progress is most likely: empirical research on human consciousness, theoretical integration, non-human systems, and interdisciplinary efforts via mathematics. We argue that empirical studies must move toward structural approaches and balance specificity with context-recognition. Further, we propose shifting from competition among the many existing theories to synthesis, clarifying their distinct explananda and integrating complementary insights. Extending research beyond humans, we recommend a modest biological naturalism for non-human animals, theory-guided approaches for artificial systems, and continued exploration of basal biological

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

systems to broaden and refine the concept of consciousness. Naturally, this leads to highlighting the importance of interdisciplinarity, such as an integration of mathematics and information theory into the field. Together, these trajectories outline an integrative, interdisciplinary vision for a science of consciousness with transformed concepts.

Keywords

Consciousness; Neural Correlates of Consciousness; Mathematical Consciousness Science; Naturalistic Research; Structuralism; Non-human Consciousness

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Declaration of interests

None.

Introduction

In earlier chapters of this volume, leading researchers have reflected on how consciousness science has evolved over recent decades. Thanks to their foundational work, consciousness is now recognized as a legitimate and scientifically tractable domain. Substantial discoveries have been made, and critical methodological and theoretical advancements have solidified its standing as a multidisciplinary research field.

As one of the closing chapters of this book, we offer a perspective from a group of early-career researchers with interdisciplinary backgrounds aiming to look ahead.¹ We identify emerging trends, methodological challenges, and conceptual shifts that may shape the future of consciousness research. Particularly, we explain why our ideas and suggestions should shape consciousness science in the years to come. The topics addressed in this chapter were selected by reorganizing and integrating themes presented across this volume. This structure allows us to continue the structure of this book integrating previous work while pointing in new directions.

We have chosen four main areas to identify and suggest trends in consciousness science that we believe will and should shape the future of our field: The empirical research of human consciousness, theory in consciousness science, consciousness in non-human systems and mathematics as an instance of the essential interdisciplinary nature of the field. Our selection inevitably comes at the expense of other possible themes. Why focus here? Because here we see investments leading to the next concrete and most productive progress for consciousness science. Other areas remain important, but we find here highly achievable leverage points likely to shift the field in decisive ways.² We continue the conversation begun in earlier chapters, integrating prior insights while opening new paths forward. We revisit the core topics explored throughout this volume not to summarize them, but to suggest how their study might evolve in the years to come.

We begin with the future of empirical research on human consciousness, before turning to theoretical developments that shape and guide such research. Next, we bring together discussions on animal models and extrapolations to other systems into a single section on

¹ Lara Bräuchle with Philosophy and Cognitive Science, Niels Gaspers with Psychology and Cognitive Science, Johannes Kleiner as Physicist, Mathematician, and Neuroscientist, and finally Zefan Zheng as Neuropsychologist.

² Of course many other areas will be relevant in a future of consciousness science. One of many worth mentioning is the area of Disorders of Consciousness. But as we see clear and direct potential for large and highly relevant and needed changes in what we are about to discuss, we limited our proposal to these four areas.

consciousness beyond the human case. Finally, we conclude with an outlook on mathematical and information-theoretic research programmes to consciousness, to showcase the importance of interdisciplinarity and the potential of mathematical approaches as a unifying interdisciplinary force.

Having just sketched the overall structure of the chapter, let us now describe more closely the first theme: the future of empirical research on human consciousness. Here, we introduce two methodological approaches that frame the debate: the specificity approach, which seeks to isolate the target of investigation (e.g., consciousness) from all other mental processes, and the context-recognition approach, which emphasizes studying consciousness as embedded within broader cognitive, bodily, and environmental contexts. Balancing both approaches is a methodological strength the field must achieve.

In the beginning part of the chapter, we provide our recommendations on the future studies of unconscious processing and neural correlates of consciousness. We recommend that future research on unconscious processing move beyond relying on simple “unconscious conditions” defined by contrasts between congruent and incongruent trials. Instead, studies should focus on identifying structural and incremental transformations from unconscious to conscious processing that will not only reveal whether unconscious processing exists, but also how unconscious processing differs from conscious processing. Similarly, research on the neural correlates of consciousness (NCC) should not rely solely on the contrastive method, which compares seen and unseen conditions. Rather, it should incorporate structural constraints that capture the relational nature of conscious content, while also striving to disentangle NCC from related cognitive functions such as attention, memory, report, and objective recognition.

We then revisit the concept of context-recognition and discuss its complementary insight. Specificity with its practical and terminological precision is needed to avoid confounding consciousness with neighbouring processes, yet by itself it risks stripping away the phenomenon itself. As a complement, Context-recognition addresses this by treating attention, memory, embodiment, and environment not as noise, but as integral dimensions of conscious experience and ensures that consciousness is not studied in a vacuum but within the lived, embodied, and ecological entanglements in which consciousness actually exists. Advances in immersive technology, portable neural measurement, and large-scale multimodal experiments now make it possible to implement this research in practice. Studies combining neural data, physiology, eye-tracking, and phenomenological reports in real-world environments will open

an unprecedented window onto conscious experience. Specificity prevents confusion, context-recognition prevents distortion.

This idea of complementation rather than competition is the core of our following proposition for theoretical developments. With over 200 theories of consciousness proposed, the field risks fragmentation (Kuhn, 2024; Seth & Bayne, 2022). Equipped with specific terminology and recognition of context, we find competing theories often target different explananda of consciousness. Progress will therefore depend less on eliminating rival accounts and more on integrating them into coherent, complementary frameworks. Emphasising synthesis over arbitration involves clarifying which aspects of consciousness each theory addresses and exploring how they might complement rather than compete with one another. Theory synthesis is a promising strategy for integrating divergent perspectives. Rather than eliminating competing accounts, synthesis aims to unify them by clarifying the specific explananda they address.³

Having established the call for structural methodology and integrative, complementary frameworks, we move on to discuss a future science of consciousness that goes beyond humans. Inquiries into non-human systems (biological and artificial) are central to the future of the field. They broaden the scope of inquiry and force us to refine what we mean by consciousness itself. We examine three major strategies—theory-heavy, theory-neutral, and theory-light—highlighting their strengths and limitations in different contexts. In animals, we push for a modest biological naturalism, while in artificial systems theory-guided approaches are more appropriate to avoid behavioral markers being gamed. Additionally, basal systems (plants, microbes, collectives, organoids), though speculative, enrich our conceptual toolkit and help clarify the phenomenon’s boundaries. Non-human doesn't have to be just an extension of human-focused science—it can be a transformation. Evaluating what consciousness means across biological and artificial spectra strengthens both ethics and theory, guiding the field toward a more integrative and inclusive science.

This integrative spirit echoes in our urging the interdisciplinarity of the field. To advance, we will likely have to fundamentally question and revise existing concepts and assumptions. The required new thinking and insight is only possible with a research spirit very open to fresh methods, terminology and ideas. New concepts and perspectives will be gained by

³ Editors' note: For one approach to theory-synthesis aiming to align different consciousness theories, see Ch. 2.7.

integrating a plurality of disciplines—and thus methods—into the field. Here again synthesis is crucial to recognize widespread common ground and complementary additions, rather than artificial competition.

To investigate what this may look like, we discuss the discipline of mathematics as already growing in relevance to the field. Here we identify three concrete trends: the emergence of a dedicated mathematical community that establishes checks and balances; a shift from merely adding new models toward improving existing ones; and the move from imported tools toward mathematics crafted for the unique challenges of consciousness. These developments will not only shape consciousness science from within, but even create opportunities for consciousness science to export innovations to other areas of science. As a final outlook, we discuss how integration of mathematics can contribute to forming new paradigms.

We conclude by imagining what a consciousness science might look like, that was transformed in its terminology, methods, concepts and paradigms by the approaches we discuss. Together, our discussions point out where the field is headed, what conceptual and methodological innovations are needed, and how a new generation of researchers might shape the science of consciousness in the decades to come.

1. Human Consciousness

Famous thinker Sigmund Freud distinguished the human mind into three parts: the conscious mind, the preconscious mind, and the unconscious mind (Freud, 1923). Despite the fact that his theories were labelled as unscientific later (Popper, 1983), the capacity of the unconscious mind and the emergence of the conscious mind have become the central topics of scientific endeavor on consciousness in the past three decades. In these efforts, modern consciousness scientists have predominantly applied something we term “specificity approach“, which aims to isolate the target of investigation and exclude all other irrelevant (mental) processes as confounds. However, there have also been philosophical concerns that consciousness is not an isolated entity but deeply embedded in its broader context. Striking this balance between specificity and context-recognition, we can foster a more comprehensive understanding of the phenomenon. Importantly, the approaches are neither new nor unique to

consciousness research.⁴ It is, however, especially salient in a young discipline like consciousness science, where our grasp of the object of investigation is still limited. The interplay of the approaches is not only inevitable—it is productive. This is what section one will explain and elaborate. First, we dive into how consciousness has been contrasted with the unconscious, and how this contrast can be drawn in a more insightful way.

1.1 The Capacity of the Unconscious Mind: From the Simple Unconscious Condition to Structural Differentiation

In everyday life, we often assume that our brains only process sensory information we are consciously aware of. However, the Freudian iceberg theory (Freud, 1923) pointed out that our mind is not predominantly occupied by the conscious mind but the unconscious one. This proposal poses many intriguing questions to consciousness scientists nowadays. Do we have an unconscious mind? Can we process information outside of our conscious awareness? If yes, which kind of information can be processed unconsciously? Is our unconscious mind really stronger than the conscious one? Therefore, consciousness scientists have extensively investigated whether sensory stimuli can be processed in the brain without reaching conscious awareness in the past decades.

Traditionally, researchers have approached this by attempting to create an "absolute unconscious condition," in which it is assumed that participants do not consciously perceive weak stimuli in any trial (Schmidt & Vorberg, 2006; Stockart et al., 2025). This assumption is

⁴ These approaches we point out are not new, nor unique to consciousness science. In fact, prominent philosophers of science and consciousness researchers have long grappled with it. Philosophers such as Nancy Cartwright and Sandra Mitchell argue that isolating phenomena through idealized models may obscure the complexity inherent in real-world systems (Cartwright, 1983; Mitchell, 2012). It also appears, for instance, in biology and history as the 'splitter' side of the lumpers–splitters distinction: splitters emphasize fine-grained differences, while lumpers stress commonalities (Endersby, 2009; McKusick, 1969; Simpson, 1945). We consider specificity essential for consciousness science, since the phenomenon is not yet clearly defined and must be distinguished from confounds such as attention. This is crucial both for empirical research and for theoretical work, ensuring that theories make explicit which aspects of consciousness they address. To name examples of both approaches in consciousness science: David Chalmers exemplified the demarcation approach by distinguishing phenomenal from functional consciousness (Chalmers, 1996), whereas phenomenologists like Francisco Varela and Evan Thompson emphasize that consciousness cannot be fully understood apart from its bodily, cognitive, and environmental context (Thompson, 2015; Varela et al., 1991). They integrate specificity in their method of neurophenomenology with more holistic frameworks. Besides these examples of the approaches in theoretical consciousness science, they also appear in empirical research, where consciousness is commonly attempted to be isolated to study (controlling for processes such as attention and memory) and at the same time its context-dependency highlighted in naturalistic experiment setting, as will be described in section 1.3. The tension between these compatible approaches is not a flaw, but a fundamental feature of inquiry into complex, context-dependent phenomena such as consciousness.

typically supported by participants' subjective reports indicating the stimulus was "invisible," either trial-by-trial (Avneon & Lamy, 2018; Koivisto & Grassini, 2018; Lin & Murray, 2014; Micher & Lamy, 2023; Trübutschek et al., 2017; Zheng et al., 2025) or within equivalent experimental conditions (Naccache et al., 2002; Naccache & Dehaene, 2001; Schmidt, 2000; Van Gaal et al., 2008; van Gaal et al., 2010; Van Gaal et al., 2010). Cognitive processing in such studies is typically measured using response priming, which refers to differences in response times to a target depending on whether the preceding prime is representationally congruent (congruent condition) or incongruent (incongruent condition) with the target. An example of a congruent condition is made of a prime being the word BLUE followed by a target in color blue. If the prime word is YELLOW while the target is blue, this is an incongruent condition. Incongruent primes delay response time to the target, which is termed prime effect. The unconscious literature leverages this phenomenon by rendering the physically displayed prime invisible using various techniques. When the response times to targets following incongruent unconscious primes are significantly longer than those of congruent unconscious primes, an inference of unconscious processing of the prime is made.

Across many studies, significant priming effects have been observed in the absence of conscious awareness of the prime, suggesting that the brain can unconsciously process a wide range of stimulus features—from low-level attributes like color (Ro et al., 2009; Schmidt, 2000) to complex ones like faces (Yang & Yeh, 2018), words (Dehaene, Naccache, et al., 1998), and even sentences (van Gaal et al., 2014).

However, this approach has faced increasing criticism. A central concern is the reliability of subjective visibility reports. Near the perceptual threshold, the boundary between visible and invisible becomes ambiguous, and participants may misreport their experiences—misreporting a weak but visible stimulus for an invisible one, or vice versa (Fahrenfort et al., 2025; Rothkirch et al., 2022; Schmidt et al., 2024; Shanks, 2017; Stockart et al., 2025). Such misclassifications introduce residual conscious awareness into trials labeled as "unconscious," potentially inflating the observed cognitive effects. As a result, many positive findings might not reflect true unconscious processing. This violates the principle of the specificity approach: isolation of the target phenomena, i.e., the unconscious processing from all other irrelevant mental processes, i.e., the conscious processing.

To overcome these limitations, we propose a shift away from the reliance on simple unconscious conditions defined by subjective visibility and binary comparisons between

congruent and incongruent trials. Instead, we advocate for a research program that focuses on identifying structural differences between unconscious and conscious processing—differences that cannot be explained by residual awareness alone.

One illustrative study by de Gardelle et al. (2011) measured response times to target faces following prime faces that varied across 11 levels of morphing between two target identities (Figure 1A). When the primes were visible, response times showed a non-linear, all-or-none relationship with prime-target similarity. In contrast, when the primes were invisible, response times varied linearly with prime-target dissimilarity. If residual awareness or misclassification of visibility were responsible for the effects in the invisible condition, the response pattern would likely mirror that of the visible condition. The observed difference instead supports a genuine structural dissociation between conscious and unconscious processing: unconscious representations are linear, whereas conscious representations are categorical.

Another example comes from Zheng et al. (2023, Figure 1B), who examined response times to color targets influenced by both bottom-up (the actual stimulus) and top-down (the participant's interpretation) representations of an ambiguous prime. When prime visibility was low, bottom-up and top-down influences were additive—suggesting separate, independent contributions. However, when visibility was high, the effects became non-additive, indicating interaction. The shift from additive to interactive effects across visibility levels reveals a transformation in cognitive architecture from unconscious to conscious processing—again, without relying on a strict unconscious condition.

A third example, illustrated in Figure 1C, is provided by Biafora and Schmidt (2020), who held the visual contrast of a masking stimulus constant while varying the stimulus onset asynchrony between prime and target. As this asynchrony increased, participants' ability to identify the prime decreased, indicating reduced conscious awareness. Simultaneously, the priming effect increased. This inverse relationship between visibility and priming strength suggests a dissociation between conscious awareness and cognitive processing, offering strong evidence for the presence of unconscious processing that does not depend on visibility-based categorization.

These results demonstrate the need for a more robust research agenda focused on uncovering structural distinctions between unconscious and conscious cognition—those that cannot be accounted for by measurement artifacts or residual awareness. Simple comparisons

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

based on visibility assumptions are insufficient. Instead, identifying transformations in processing architecture offers deeper insight into the nature of conscious and unconscious mind. Adjusting empirical research of human consciousness to this fact, is the first suggestion we propose for the future science of consciousness.

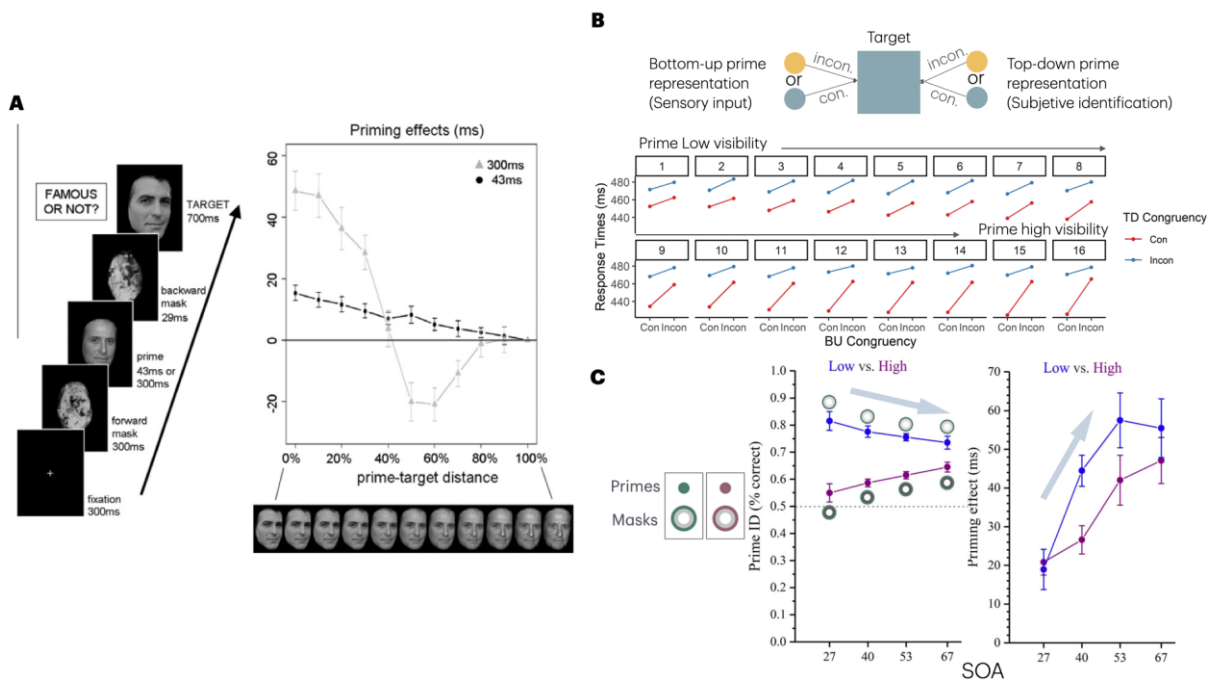


Figure 1. Structural dissociations between unconscious and conscious processing. (A) *Linear vs. nonlinear priming effects:* When participants were presented with morphing face primes of varying similarity to the target faces, response times increased linearly with dissimilarity under low visibility (grey line), but showed an all-or-none, step-like pattern when primes were consciously visible (black line). (B) *Additive vs. interactive effects of top-down and bottom-up representations:* Under low prime visibility, bottom-up (physical) and top-down (interpreted) representations produced additive priming effects on color judgments. Under high visibility, the effects became under-additive, indicating integration during conscious processing. (C) *Dissociation between visibility and priming strength:* Increasing the prime-target stimulus onset asynchrony (SOA) while holding mask contrast constant reduced prime visibility (blue arrow left) but enhanced priming effects (blue arrow right), revealing a dissociation between awareness and cognitive processing strength. Figure adapted from Biafora and Schmidt (2020), de Gardelle et al. (2011), and Zheng et al. (2023) after permission.

1.2 Neural Correlates of Consciousness: From Contrastive Method to Structuralistic Constraints and Beyond

Besides the capacity of the unconscious mind, a central pursuit in consciousness science is understanding how the conscious mind emerges from brain activity. One influential empirical strategy is the search for neural correlates of consciousness (NCCs) — the minimal neural activities that are sufficient for the generation of conscious experiences (Crick & Koch, 1990).⁵ A contrastive method has been applied to isolate NCCs in empirical investigations, which contrasts the brain activities with conscious experience of a sensory input with those without. This method has yielded substantial findings, often implicating widespread cortical activity—including prefrontal regions—and has provided empirical support for theories such as the Global Neuronal Workspace (GNW) (Dehaene et al., 2001; King et al., 2016; Salti et al., 2015; Sergent et al., 2005; Van Vugt et al., 2018).⁶

However, a central critique of this contrastive paradigm is its lack of content specificity (Lepauvre & Melloni, 2021). While it may reveal that consciousness is present, it often fails to explain how the brain represents particular qualitative features of experience—such as what makes red feel different from blue, or why red feels more similar to purple than to green. Understanding the phenomenal character of experience—what it is like to see red, for instance—remains a fundamental goal of consciousness science (Kawakita et al., 2023). This goal cannot be met by merely identifying the presence or absence of awareness; it requires more fine-grained methods capable of capturing the structure and richness of conscious content.

Recent insights from structuralist philosophy help sharpen the focus (Fink et al., 2021; Kleiner, 2024; Malach, 2021). As detailed in the chapter on Structuralism and Neural Correlates of Consciousness, structuralist thinkers argue that the phenomenal character of an experience is entirely defined by its relations to other experiences (Ladyman, 1998; Worrall, 1989).⁷ There is nothing intrinsically unique about "redness"; rather, its identity comes from its place within a web of similarities and differences among all other colors. Extending this

⁵ Editors' note: For a broader discussion on challenges with the NCC approach, see Ch. 1.2.

⁶ Editors' note: For further information on the Global Neuronal Workspace framework and a perspective on its relationship to conscious access, see Ch. 2.1.

⁷ Editors' note: For an elaboration on how one might utilise structuralism to address the challenge of phenomenal character, see Ch. 1.6.

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

view to neuroscience, structuralists propose that the neural substrates underlying conscious experience must mirror the relational structure of phenomenological space. For example, the neural representation of red should be more similar to that of purple than to that of blue, just as these relationships hold in our subjective experience.

This idea introduces a structural constraint on the search for NCC: rather than merely identifying brain regions activated during conscious perception compared to unconscious perception, researchers should also ask whether the geometry of neural representations reflects the structure of phenomenological experience (Fink et al., 2021). Importantly, one need not fully subscribe to structuralist philosophy to apply these constraints—only to accept that structural correspondence between neural and experiential spaces is relevant for isolating true NCC.

However, an important caveat must be acknowledged: structural correspondence alone is not sufficient to generate conscious experiences. Empirical studies have shown that the neural representational geometry of many perceptual categories—such as spatial layouts (Vorberg et al., 2004), faces (de Gardelle et al., 2011), numbers (Hesselmann et al., 2015; Koechlin et al., 1999), and objects (Haberkamp et al., 2019)—can be similar in both conscious and unconscious states. This indicates that unconscious representations may also preserve the same structural relationships as the conscious ones in the brain. Thus, while the structural constraint is a valuable tool for identifying potential NCC, it must be applied across both conscious and unconscious conditions to rule out confounds from unconscious representations in future consciousness research.

Returning to the aforementioned first critique of NCC research: In addition to concerns on content specificity, the contrastive method itself is also vulnerable to cognitive confounds. When comparing "seeing" versus "not seeing" conditions, differences in neural activity may reflect more than just consciousness. Participants who see a near-threshold stimulus are also more likely to attend to it, retain it in working memory, make decisions about it, and generate verbal reports (Aru et al., 2012; Lepauvre & Melloni, 2021). Consequently, the contrastive method risks conflating NCC with the neural correlates of attention, memory, decision-making, and motor responses.

Effectively separating NCC from these cognitive processes requires demonstrations that consciousness can be dissociated from them on a functional level—a challenge that remains

unsettled. Some theoretical accounts, such as the GNW theory, argue that conscious experience is inherently tied to attention, working memory, and decision-making (Cohen et al., 2016; Cohen & Dennett, 2011; Kouider et al., 2010). This view characterizes consciousness in terms of access: we are conscious of what we can attend to, manipulate, and report. In contrast, other frameworks—such as Recurrent Processing Theory (RPT) and Integrated Information Theory (IIT)⁸—maintain that phenomenal consciousness is richer than what can be accessed or reported, and that it may arise independently of higher-order cognitive functions (Block, 2011; Fu et al., 2021; Sligte et al., 2010).

If GNW is correct, then isolating NCC from attention and memory is not just empirically difficult—it is conceptually misguided. If RPT or IIT is correct, then dissociating consciousness from cognition is both theoretically meaningful and methodologically necessary. Therefore, advancing NCC research will depend on more than refining our experimental tools; it will require a deeper understanding of how consciousness relates to cognitive systems.

Following this approach, we argue that the future of NCC research should move beyond the contrastive method and incorporate structural constraints on both conscious and unconscious conditions that account for the relational nature of only the conscious content, and efforts must be made to disentangle consciousness from cognition on both functional and theoretical grounds.

The typical rationale of this NCC program falls under the realm of the specificity approach: it commonly attempts a clear demarcation of consciousness from other mental faculties and rigorous experimental control, trying to ensure that the neural patterns identified reflect consciousness itself, rather than related functions that are potentially irrelevant to consciousness.

1.3 The Context-Recognition Approach

The previous sections outlined the advantages and rationale behind the specificity approach, which investigates consciousness by carefully demarcating it from unconscious processes and potential confounding factors such as attention, memory, and verbal reports (see Block, 2019 as an example). We suggested advancing this approach by focusing on

⁸ Editors' note: For further reading on Integrated Information Theory and its axioms, see Ch. 2.5.

structural aspects. However, many researchers highlight a complementary path: instead of treating these “confounding factors” as noise to be controlled for, we might productively understand them as part of consciousness’s broader context and investigate how they relate. As previously discussed, ongoing debates (such as whether attention is necessary for consciousness) demonstrate how intertwined these domains remain.

And the entanglements do not stop at cognition. Arising discussions around non-standard forms of consciousness (such as in infants, animals, plants, or artificial systems) highlight the need to consider evolutionary and biological embeddedness. At this early stage in consciousness science, we lack the conceptual and empirical tools to isolate consciousness from these embedding factors without risking distortion of the very phenomenon we seek to understand. Correlations with biology, embodiment, and cognition do not in themselves imply necessity or sufficiency for consciousness, they are, however, empirically robust. Ignoring these entanglements risks overlooking essential dimensions of conscious experience—something we cannot afford in such a young, still-developing science. Context-recognition should not replace specificity, but opens up complementary questions: How does experience vary across bodily states, meditative and altered states of consciousness, developmental stages, or ecological niches?

Philosophical traditions like Phenomenology have long emphasized the situated nature of experience. The 4E approach (embodied, embedded, enactive, and extended cognition) continues this line, asserting that consciousness cannot be fully understood apart from its bodily and environmental context (Newen et al., 2018). Recognition of these entanglements has shown its potential to reveal previously overlooked dependencies and shift understanding of conscious experience.⁹ As discussed previously, we currently lack the conceptual and empirical tools to cleanly separate consciousness entirely from its entanglements. A more productive approach might thus be to reflect on the starting point of where we are investigating consciousness from: being humans with body and mind embedded into a (social) environment. Recognising this opens up a perspective to imagine what consciousness could mean outside a neurotypical human framework and consider its entanglements.

⁹ Take for example Merleau-Ponty’s body schema, in which he showed already in 1945 that phenomena such as typing cannot be captured with at the time prominent behaviourist frameworks, but require the entire schema of the body (Merleau-Ponty, 2013). This could later be adequately captured by cognitive science’s understanding of complex cognitive-motor processes.

Experimentally, this has led to the development of research programs that go beyond highly controlled, contrastive paradigms. Frameworks such as neurophenomenology (Varela, 1996), micro-phenomenological interviews (Petitmengin et al., 2019; Valenzuela-Moguillansky & Vásquez-Rosati, 2019), and neuroethology (Cabral-Calderin et al., 2025) actively engage the lived, embodied, embedded or dynamic character of conscious experience. These methods investigate experience as it unfolds in ecologically valid environments, rather than stripping it of its complex context for the sake of control. And the evidence is compelling: naturalistic experiments produce neural and behavioral signatures that differ significantly from those observed under constrained laboratory conditions.¹⁰

Recent technological advances further enhance the potential of the context-recognition approach. Advances in computing power, data storage, and immersive technologies are now enabling a new generation of empirical research aligned with the approach (e.g., Hirschhorn et al., 2024). One important development that will shape consciousness research is how portable neural and physiological measurement tools allow for the investigation of brain, behavior, and cognition in increasingly naturalistic and previously less-explored settings (see Lewis-Healey et al., 2024). These possibilities give new relevance to fields such as neurophenomenology and Neuroethology. Consciousness research is beginning to explore these possibilities for experiments with a much higher degree of ecological validity. Interpreting data from naturalistic settings, however, requires theoretical models capable of accounting for an embedded and dynamic nature of conscious phenomena. In this sense, technological progress is not merely extending the empirical realm of consciousness science, but it is reactivating philosophical concerns about relationality, embodiment, and context-dependence.

The interplay between the specificity and context-recognition approach forms a generative dynamic where each offers complementary insights: specificity sharpens experimental control and conceptual clarity, while context-recognition broadens the scope of inquiry, enriching it with ecologically valid data and relational perspectives. A more complete understanding of consciousness arises not from their competition, but from their integration. Many

¹⁰ Mudrik et al. (2024) gathered several excellent examples of research to show how context turned out to make a great difference. They mention for example how “the widely established face-related N170 component was found in EEG studies for the 2D stimuli, it was abolished in response to 3D ones”, referencing the study by Sagehorn et al. (2023).

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

contemporary theories and research programs reflect this integration¹¹—describing the approaches is not meant to categorize research, but rather to illuminate the methodological and conceptual commitments that shape strategies. This awareness helps clarify where traditions originate, how they can complement one another, and how some debates may be influenced by differing assumptions about whether to emphasize isolation or embeddedness.

The advancing of experimental design, theoretical reflection, and technological capacities increasingly equips researchers to both put their finger specifically on consciousness and investigate it within its lived, embodied, and ecological context. Better terminology and precise experimental setups are just as beneficial as conceptual revisions informed by contextual entanglements. Recognizing and articulating the productive differences between specificity and context-recognition can be a helpful framework for understanding research in the field. The scientific investigation of consciousness is driven by both efforts: the demarcation of its boundaries, and the exploration of its dependencies.

Section 1 set out a new agenda for empirical research on consciousness. We argued that methodological approaches must be rethought: research should establish how conscious processing differs structurally from unconscious processing. Likewise, work on neural correlates of consciousness (NCCs) should focus on structural constraints that capture the relational nature of conscious content. This will be the new methodological key in important research to achieve specificity: avoiding confounds with other mental processes or neural activity.

Yet specificity alone is not enough. Consciousness is deeply context-dependent, and a science that isolates it too much risks stripping away the very phenomenon it seeks to explain. The context-recognition approach highlights these entanglements. With advances in technology and interdisciplinary collaboration, the context of conscious experience can be increasingly considered and explicitly studied. Large, integrative experiments combining neural data, eye-tracking, physiology, and real-world settings are increasingly possible and would offer fundamental new insights for the field as a whole.

¹¹ Most research objectives show aspects of both approaches. However, they commonly highlight one over the other, which can lead to misunderstandings of the research objective between disciplines or teams. Making them explicit and recognizing them in several research agendas can help the field to better communication and more productive research.

Balancing both approaches makes the field methodologically stronger: specificity prevents confusion, while context-recognition guards against artificial isolation. Theoretical work also benefits: clarity about which aspect of consciousness a theory addresses can foster complementarity rather than competition. The next section explores this idea.

2. Theorizing Consciousness: From Theory Arbitration to Theory Synthesis

In the previous sections, we discussed the debate on whether consciousness can be functionally dissociated from cognitive processes such as attention, working memory, and verbal report. And despite two decades of empirical investigation, the debate remains unresolved. Both sides have accumulated substantial empirical support, yet neither has been definitively challenged (Chen & Wyble, 2015; Cohen et al., 2014, 2015, 2020, 2020; Landman et al., 2003; Pinto et al., 2013; Sligte et al., 2010; R. Wang et al., 2021). This stalemate is partly due to the alignment between experimental paradigms and theoretical expectations: researchers tend to design studies that favor their preferred framework and interpret results accordingly, often as confirmation rather than contradiction (Yaron et al., 2022). As a result, incompatible theories persist side by side.

This is not an isolated case. GNW and IIT/RPT camps have failed to settle many more debates against each other in the past decades (Dembski et al., 2021; Naccache et al., 2016; Overgaard et al., 2006; Sergent & Dehaene, 2004). The field of consciousness research is replete with such unresolved debates. In fact, some reviews estimate that over 200 theories of consciousness currently circulate in the scientific literature (Kuhn, 2024; Seth & Bayne, 2022). There are fierce debates going on between many of the theories besides those between GNW and IIT/RPT (Corcoran et al., 2025; Maniscalco & Lau, 2016; Michel & Lau, 2021; Phillips, 2021; Schwiedrzik et al., 2011). If consciousness has a singular mechanistic explanation, this proliferation of theories is unsustainable. Yet, if a single debate between just two or three theories resists resolution, how can the field hope to converge on one unifying explanation between 200 theories?

In response to this challenge, the COGITATE project initiated a high-profile attempt at theory arbitration (Ferrante et al., 2025; Melloni et al., 2021, 2023). Using the method of

adversarial collaboration proponents of GNW and IIT jointly agreed upon an experimental design while making opposing predictions.¹² The intention was clear: if the results matched one theory but not the other, the latter would be falsified, and its proponents would concede. This transparent and pre-registered approach held great promise for resolving theoretical disputes.

Yet, the outcome was sobering. While the results posed challenges to both GNW and IIT, neither side altered its theoretical commitments (Ferrante et al., 2025). Several explanations have been offered for this impasse. Some cite methodological limitations in the experimental design that prevented clear interpretation (Lau, 2023; Lepauvre, 2025). Others point to the inherent difficulty of overturning entrenched theoretical positions—a challenge even acknowledged by Daniel Kahneman, who championed adversarial collaborations and noted that such efforts rarely lead to “changed minds” (Kahneman, 2003).

If even the most rigorous and transparent arbitration efforts cannot dislodge theories from the landscape, what alternative path remains? We propose that the root of the problem may lie in a fundamental oversight: many theories of consciousness may be addressing different explananda.

Returning to the example above, GNW theorists argue that consciousness—specifically access consciousness—emerges when information is globally broadcast from local sensory regions to the prefrontal cortex, making it available for attention, memory, reasoning, and report (Dehaene, Kerszberg, et al., 1998; Mashour et al., 2020). In contrast, IIT and RPT posit that phenomenal consciousness arises within local sensory areas through recurrent interactions, and that prefrontal involvement reflects subsequent cognitive processing rather than conscious experience itself (Fu et al., 2021; Tononi et al., 2016). Consequently, GNW experiments often involve directed attention and explicit reports, while IIT and RPT studies frequently use no-report paradigms or divert attention away from the stimuli (Yaron et al., 2022).

This divergence suggests that the ongoing debates may not stem from empirical contradictions, but from theoretical misalignment—different theories might be describing different facets of a multifaceted phenomenon (Northoff & Lamme, 2020; Signorelli et al.,

¹²Editors' note: For further background on the concept of adversarial collaboration, such as COGITATE, and how it may be operationalized in the research of consciousness, see Ch. 2.7.

2021; Storm et al., 2024). If consciousness comprises multiple components, such as access and phenomenality, then theories like GNW and IIT may both be valid within their respective domains. When clear boundaries are drawn around these explananda, apparent contradictions may dissolve, and the need for arbitration diminishes.

In this context, a more promising approach to the "too-many-theories" problem is not elimination, but synthesis (Fazekas et al., 2024; Luppi et al., 2024; Northoff & Lamme, 2020; Singhal & Srinivasan, 2024; Storm et al., 2024; Whyte & Smith, 2021; Zheng et al., 2025). Theory synthesis involves integrating multiple theories—each explaining different aspects of consciousness—into a coherent framework. This process requires identifying which explananda each theory addresses, resolving points of incompatibility, and articulating a unifying structure that is more than a mere sum of its parts.

Importantly, theory synthesis is not simply an exercise in philosophical accommodation. It must also involve integrating the empirical evidence that supports each theory. Recent efforts in this direction include aligning theories based on neurobiological compatibility (Storm et al., 2024), comparing them based on explanatory strategies (Signorelli et al., 2021), or organizing them according to the temporal and spatial scales of neural activity they emphasize (Northoff & Zilio, 2022). While these are valuable steps, we argue that true synthesis must go further. It should not treat theories like modular "Lego pieces," but instead identify shared principles or common threads that transcend individual accounts (see Zheng et al., 2025, Lopez & Wiese, 2025). These unifying elements can serve as the foundation for a deeper framework—one that reveals the minimal architecture necessary for consciousness to arise.

To recap: the first two sections lay the groundwork for the chapter's central aim. Methodological progress requires structural approaches and balancing specificity with context-recognition, while theoretical progress builds on this, demanding a move beyond arbitration toward synthesis. Together these shifts push the field toward a more integrative science—one that avoids false isolation, fosters complementarity, and opens the way to a coherent framework. Ultimately, such a framework should also extend beyond human consciousness to incorporate the possibility of conscious experience in non-human animals, artificial systems, and other underexplored domains. In doing so, theory synthesis not only addresses the fragmentation of current theory but also broadens the scope of inquiry, guiding the next generation of consciousness research toward greater coherence and inclusivity.

3. Consciousness Beyond the Human Case

Inquiries into consciousness in non-human systems have been a part of the cognitive sciences and philosophy for several decades. While early works in cognitive science probed the possibility of creating thinking machines, rather than conscious machines explicitly, the prevailing computational functionalism opened the door for the possibility of machine consciousness (Neumann & Kurzweil, 2012; Newell et al., 1958; D. Thompson, 1965; Turing, 1950).

While some prominent criticisms against this wave of computational functionalism followed (Searle, 1980), emerging theories of consciousness (e.g. GWT) offered frameworks for a more rigorous approach to machine consciousness. Similarly, questions about the inner life of non-human animals have been with philosophers and scientists at least since Descartes (Duncan, 2006). After behaviourism fell out of favour, interest in non-human animal's minds was rejuvenated, with increased focus on the ethical implications of animal sentience (Gallup, 1970; Griffin, 1977, 1981; Nagel, 1974; Premack & Woodruff, 1978; Singer, 2023). The range of candidate animals for consciousness has continuously expanded to include most vertebrates, as well as some invertebrates¹³ (Duncan, 2006; Wickens, 2022). The study of consciousness has expanded even further to include the possibility of minimal awareness in other biological and artificial systems such as plants, microbes, collective organisms (e.g., ant colonies), or lab-grown neural organoids¹⁴.

Moving forward, it will remain an important task to refine and advance our understanding of both artificial and non-human animal consciousness. For one, the presence or absence of consciousness in a system guides ethical judgments and legislation. Therefore it is vital to ground these decisions with proper evidence, as to not unjustly dismiss conscious systems

¹³ Prominent examples include cephalopods (Ponte et al., 2022), decapod crustaceans such as true crabs and lobsters (Crump et al., 2022) as well as several insect species (Barron & Klein, 2016; Chittka, 2017; Overgaard, 2021).

¹⁴ There is no room for detailed discussions of neural organoids in the space of this chapter, but see Gaillard, 2024; Lavazza & Massimini, 2018; Niikawa et al., 2022; Owen et al., 2024. Since ethical considerations seem pertinent in discussions of consciousness in cerebral organoids and since it is hard to settle questions about their consciousness status, see Kammerer (2022) for ethics without consciousness and Dung (2024) for an opposing view. See also Birch (2024), Dung (2023b, 2023a) and Elbro (2024) for further discussions of consciousness and ethical ramifications.

Editor's note: see also Ch. 6.2 „Perspectives on Consciousness in Human Brain Organoids“ in this volume, where Andrea Lavazza reviews recent advances in organoid technology and theoretical considerations on the possibility for consciousness in neural organoids.

from these considerations. Another reason for the importance of non-human consciousness research lies in the prospects that understanding the distinction between systems that possess consciousness and those that lack this property is very likely to advance consciousness research as a whole. Additionally, it enables the comparison of the concept of consciousness across systems that may instantiate it in ways that are not yet captured by prominent theories.

While non-human consciousness has been studied for several decades, recent developments are driven by a rigorous theorising of consciousness. Current investigations can be broadly grouped into three methodological approaches.

Theory-heavy approaches assess whether a system meets the architectural or functional requirements of formal theories of consciousness, such as GWT, IIT, or Recurrent Processing Theory (RPT). For example, GWT proposes that conscious perception involves representations in a neural or cognitive module that broadcast information across the system's cognitive architecture. The presence of such a mechanism is taken as evidence for consciousness. The goal is to determine whether non-human systems exhibit the structural and functional capacities that theories of consciousness designate as necessary for conscious experience.

At the other end of the spectrum, theory-neutral approaches focus on identifying behavioral signatures associated with conscious processing in humans. These approaches apply broadly, assuming systems are capable of comparable behavior, but rely heavily on heuristics. This makes them prone to over-attributing consciousness to human-like systems and underestimating it in systems that are physiologically or perceptually dissimilar, especially disembodied systems.

Theory-light approaches strike a middle ground, examining the co-occurrence of cognitive traits associated with consciousness, such as flexible learning or metacognition. In characterising these approaches, Birch (2022) stresses the importance of species-appropriate experiments that distinguish conscious from non-conscious processing.¹⁵ This method rests on a minimal assumption:

¹⁵ Editor's note: For how structural approaches might be helpful here, see Ch 1.6.

"Phenomenally conscious perception of a stimulus facilitates, relative to unconscious perception, a cluster of cognitive abilities in relation to that stimulus." (Birch, 2022, p. 140)

This minimal assumption falls in line with recent approaches emphasizing the relation of consciousness to other cognitive capacities, as well as the embodiment of the specific system instantiating consciousness (see 1.2). Instead of relying on anthropocentric theories, theory-light approaches elucidate the “same-but-different” nature of consciousness and consciousness-related cognition and behaviour across vastly different species (Cao, 2022; Polger, 2016). While no single method suits all inquiries, theory-light and theory-heavy approaches each have advantages for specific systems.

In non-human animal research, theory-heavy methods face the specificity problem (Shevlin, 2021; see also Andrews & Delmore, this volume (Ch. 4.1: “Framing the question of animal consciousness”; Jablonka & Ginsburg, this volume (Ch. 2.6: “Learning and the evolution of consciousness”); and Wilke & Kingir, this volume (Ch. 4.2: “Seeing beyond the human: Challenges and advances in animal studies of visual consciousness”). Theories often rely on interpretations referencing human-specific cognitive mechanisms, such as metacognition, and introspective awareness of consciousness itself, making cross-species application difficult. As a result, consciousness in non-human animals is commonly explored through theory-light and theory-neutral approaches (Birch, 2022; Tye, 2017). Since biological systems are products of evolution, assumptions of evolutionary continuity give at least some grounds for assessments of consciousness via behavioral and structural markers¹⁶. Thus, theory-light and -neutral approaches, which focus on observable behavior rather than internal mechanisms, are preferable.

These approaches begin with intuitive indicators of consciousness and then seek neurobiological, behavioral, and structural correlates. One regularly used indicator of this kind is pain and pain related behaviours, like flinching and protection of damaged body parts (Tye, 2017). However, the continuously expanding list of markers these approaches produce, increases the risk of over-ascribing consciousness across systems (Andrews, 2024). Instead of addressing the distribution question (i.e. which non-human systems are conscious), marker approaches might be better used to explore different dimensions of consciousness in non-

¹⁶ While this reasoning is intuitive it is not settled at what exact point in evolutionary history a transition from non-conscious to conscious organisms could have happened. For more elaborate discussions see Birch et al. (2020), Bronfman et al. (2016), Gusev et al. (2025) and Godfrey-Smith (2016b, 2021).

human animals. We suggest extending this framework to all biological systems. While attributing consciousness to plants, protozoa, or bacteria may seem overly liberal, dismissing them without evaluation is equally unfounded. This perspective aligns with the trend of biological naturalism (an approach strong in context-recognition; see section 1.3), which posits that consciousness is deeply tied to the complex biological organization of living systems (Searle, 1980; A. K. Seth, 2025).

This trend further fits with the basal cognition research framework that has gained momentum in recent years (Barandiaran & Moreno, 2006; Godfrey-Smith, 2016a; Jékely et al., 2021; Keijzer, 2015a; Lyon, 2006; Sims, 2024; Van Duijn et al., 2006). While far from being a unified approach, basal cognition research commonly frames cognition in biological terms and emphasizes the central relevance of biological constraints and demands. The aim of this research approach is to enable understanding of cognitive processes in evolutionary late-comers, like humans, by first understanding organisms that branched off from our phylogenetic line earlier than other mammals for example (Lyon et al., 2021).

Within this research framework, even further-reaching views on consciousness emerge along the lines of biological naturalism. Proponents of plant consciousness such as Segundo-Ortin and Calvo (2022) argue that plants display complex behaviors like chemical communication, kin recognition, and adaptive foraging, supported by functionally analogous systems to animal nervous networks.¹⁷ However, critics point to methodological flaws¹⁸ (see also Calvo Garzón & Keijzer, 2011). Baciadonna and colleagues (2023) conducted a replicability review and found insufficient evidence for associative learning or sentience in plants. This debate extends to simpler organisms. Baluška and Reber (2019) argue for unicellular sentience, but critics challenge the empirical grounding of the Cellular Basis of Consciousness (CBC) theory (Baciadonna et al., 2023; Hansen, 2024).

While bacteria and protists show biochemical complexity and basic learning, inferring phenomenal experience from these traits remains speculative. Without robust cognitive markers, attributing consciousness to basal life forms remains unsupported. A survey of 166 researchers found only a minority believed plants possess consciousness (Francken et al., 2022). A general trend towards recognising the connection between mind and life, however, is

¹⁷Editors' note: For a detailed account of the discussion on the possibility of plant consciousness, see Ch. 6.4.

¹⁸ Similar problems abound throughout non-neural cognition research, see Boisseau et al. (2016); Dussutour (2021); Gershman et al. (2021); see also Shapiro (2021) for an optimistic perspective.

present in several recent works (Baluška et al., 2025; Baluška & Reber, 2019; Birch et al., 2020; Godfrey-Smith, 2019; Jablonka & Ginsburg, 2022; Jékely et al., 2021; Keijzer, 2015b; Lyon, 2006; Zacks & Jablonka, 2023).

Moving forward, research on consciousness in biological systems needs to develop beyond paradigmatic tests of specific cognitive abilities towards holistic and ecologically valid methodologies—context-recognition is particularly relevant for research in biological systems. This trend can be observed in comparative cognition and non-neural cognition research and will likely enable consciousness science to move beyond anthropocentric assumptions that lead to dismissal of a host of organisms as conscious beings (Birch, 2022; Carruthers, 2019; Standish, 2013).

Recent advancements in artificial intelligence, as well as this increased interest in basal biological systems, has ignited a debate between computational functionalism and biological accounts of consciousness. While markers of consciousness are useful for assessments of biological systems, they face several problems when employed for artificial systems. The core problem is that markers of consciousness are typically derived from observations of human or non-human animal organisms.

When attempting to apply these markers to non-biological artificial systems, researchers encounter an "epistemic wall": Markers gathered from biological systems cannot simply be extrapolated to artificial systems (McClelland, 2024). Theory-heavy approaches are better suited for artificial systems, as most theories are rooted in computational functionalism. These approaches avoid the facilitation hypothesis, which may not apply to artificial systems lacking evolutionary pressures.

Moreover, theory-neutral tests are particularly vulnerable to misuse in AI contexts. As argued in several works, artificial systems can be designed to pass specific tests of consciousness while lacking capabilities in every other domain, this is known as the gaming problem (Dung, 2025). Contemporary large language models, for instance, present a high sophistication in terms of natural language production, while simultaneously failing to show similar performance in other domains, that might be necessary for consciousness to be instantiated (Dziri et al., 2023; Lewis-Healey et al., 2024; Saxena et al., 2025); meaning theory-light and neutral approaches are ill-suited for probing consciousness in artificial systems (Andrews, 2024; Birch, 2022; Butlin et al., 2023; Blum & Blum, this volume (Ch.

5.2: “A theoretical computer science lens on consciousness”); Elmoznino et al., this volume (Ch. 6.1: “Consciousness in Artificial Intelligence”); but see Dung, 2023b for a different perspective).

A pragmatic strategy is to draw from multiple theories and extract shared computational or cognitive mechanisms that may serve as tentative indicators of artificial consciousness (Butlin et al., 2023; Rouleau & Levin, 2025). This approach taken by Butlin et al. reflects the kind of theory synthesis we see as a promising approach moving forward (see Ch. 2.7). By distilling criteria from theories like RPT and GWT, a list of indicator properties can be synthesised, which enables probing artificial consciousness with a sufficient degree of credence. While imperfect, this approach currently offers the most promising framework for addressing the question, a matter of increasing importance in both public discourse and academic research (Francken et al., 2022). Ultimately, a conceptual separation between the study of biological and artificial consciousness is essential to produce meaningful insights for each domain (McClelland, 2024).

As such, it will be beneficial to conceptually and methodologically distinguish between the study of machine consciousness and that of biological consciousness. The latter is deeply rooted in organisms' intrinsic need to navigate complex, multidimensional environments for survival and reproduction, pressures that artificial systems do not currently face. Unless artificial systems are designed under similar constraints, meaningful comparisons between artificial and biological consciousness may remain elusive. Furthermore, computational functionalism may fall short when applied to systems that do not readily conform to its assumptions. In particular, it may lead to overlooking the fine-grained, embodied features that make living systems cognitive or conscious. Recent literature on artificial consciousness also highlights a troubling trend: the outright dismissal of consciousness in biological systems such as plants or unicellular organisms based on factors like slower signaling speeds or differences in chemical transmission mechanisms (Hansen, 2024). This stance is particularly puzzling given the relatively uncritical acceptance of functional equivalence between human cognitive modules and artificial systems, such as the comparisons drawn between large language models and human semantic encoding (Katlowitz et al., 2025; Sun et al., 2024; Wang et al., 2025). This suggests not only an anthropocentric bias but also a computational one. In light of this, the ongoing reconsideration of computational functionalism, as argued by Seth (2025), deserves serious attention. We recommend adapting a modest biological

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

naturalism to properly study non-human consciousness in biological systems, while simultaneously keeping with a theory-guided and critical approach when concerned with artificial systems.

A modest biological naturalism should not presuppose a strong life-mind continuity or identity. It should, however, start from a baseline assumption of consciousness being to some degree present in non-human animals and, as recommended by Andrews (2024), focus on outlining the varieties of conscious states different animals can go through. For this approach to bear fruits it further needs to employ a methodology that assesses behavioural markers in light of ecological and bodily constraints specific to the animal that is being investigated. Only in this way will it be possible to accurately assess the range of experiences that potentially differ in many ways from our own, familiar, experiences.

This approach differs from the biological naturalism proposed by Seth (2025), in that it does not assume that biological substrate is a necessary condition for consciousness. Instead of making metaphysical claims about consciousness, it endorses a methodology that assumes biological properties as the basis of consciousness in biological systems, while also being open to different properties being the basis for consciousness in vastly different systems. Consciousness in artificial systems should not be disregarded based on assumptions of biological naturalism the same way it should not be taken for granted based on overconfidence in computational functionalism. A modest biological naturalism recognizes the importance of biological properties for consciousness in living systems while holding out judgement on artificial systems, in order to probe consciousness in all its possible instantiations.

This way, the edges of consciousness, both at the lower levels of biological organisation, as well as at the frontier of artificial intelligence can inform and improve a mature concept of consciousness. By holding off on metaphysical claims, consciousness science will be able to explore consciousness more freely. This, in turn, will lead to a broad variety of empirical data that should be utilised to refine theories and concepts and allow for more informed metaphysical claims as a result.

In this third part, we have argued that the study of consciousness must move decisively beyond the human case. Just as methodological progress depends on integrating structural approaches to achieve specificity while balancing with context-recognition, and theoretical

progress on moving from arbitration to synthesis, progress across systems requires conceptual clarity and methodological pluralism. A modest biological naturalism for non-human animals, paired with critical theory-guided approaches for artificial systems, provides the best chance of advancing without either overextending or prematurely narrowing the scope of inquiry. Taken together, the first three parts show how consciousness research can avoid isolation, fragmentation, and anthropocentrism—charting instead a trajectory toward an integrative, inclusive science of consciousness. The next section develops the required integrative spirit of the field even further. We highlight the essential need for interdisciplinarity by describing how mathematical and information-theoretic tools can strengthen the previously outlined trajectory.

4. Mathematical Approaches

Having discussed trajectories and trends in philosophy and empirical research, for the final section, we take a look at an entirely different research methodology that is starting to become relevant for consciousness science: that of mathematics.

Here, we use the term “mathematics” to denote the method, not the discipline. The new mathematical contributions in consciousness science span all fields and disciplines that make use of mathematical research, ranging from pure and applied mathematics, to physics, computer science, and the many mathematical sub-fields of the empirical sciences. We think that all of those have a crucial role to play in the novel phase of consciousness science that might be about to emerge.

Problems and questions that are amenable to or require mathematical solutions abound in most disciplines as they start to blossom and become more mature. Consciousness science is no exception. Correspondingly, formal and mathematical approaches are applied widely over various research areas in consciousness science. Research ranges from improving, extending, and constructing models and theories of consciousness (Barrett & Mediano, 2019; Blum & Blum, 2023; Down & Mediano, 2023; Kleiner & Tull, 2021; Mediano et al., 2022; Prentner & Hoffman, 2024; Signorelli et al., 2021; Tull et al., 2023), to work on measurement (Kleiner, 2025a), modelling experiments (Hanson & Walker, 2019; Kleiner & Hoel, 2021), to AI consciousness (Blum & Blum, 2020; Kleiner & Ludwig, 2024a; A. K. Seth, 2025).

Similarly, information-based methodologies are playing an increasingly important role in consciousness research. Not only do major theories—such as Integrated Information Theory (IIT), discussed earlier in the chapter—center on a notion of information, but empirical work is also following suit. Neural measures of complexity, such as Lempel-Ziv complexity, Kolmogorov complexity, and Permutation Entropy, as well as other information-theoretic analyses such as Perturbational Complexity Index (PCI), weighted Symbolic Mutual Information (wSMI), or Partial Information Decomposition (PID), are gaining traction as tools for probing consciousness (Casali et al., 2013; Down & Mediano, 2023; Lewis-Healey et al., 2024). Relatedly, brain criticality and measures of it are gaining importance in the empirical study of consciousness (Maschke et al., 2024).

Because a detailed review of some of the activities in this part of consciousness science is already provided in the chapter on Mathematical Approaches in the Scientific Study of Consciousness (Ch. 5.1) of this volume, our goal in this section is to review what we think are important trends and methodological shifts in the mathematical research landscape itself. We think that these trends and shifts will make a large difference on mathematical research, and that they will shape consciousness science in the years to come.

Overall, we hope that this chapter serves to illustrate even further the importance of interdisciplinary research in consciousness science.

4.1 Checks-and-Balances and an Emerging Community

Mathematical research was part of consciousness science from the start. Early examples include the contrastive approach to finding NCCs, first proposed by Baars (1986) and Crick and Koch (1990), whose statistical implementation relies on mathematics developed much earlier, and early work on quality spaces that draws on the mathematics of psychometrics (Clark, 1993). Slightly later examples include the mathematical formulation of IIT (Tononi, 2004), early approaches to formalize GNW (Wallace, 2005), mathematical models used as part of research in neurophenomenology (Lutz et al., 2002), and dynamical systems theory (Haken & Haken, 1996). The appearance of mathematics so early in the field is not surprising as mathematization is generally a consequence of progress in a scientific discipline (Ferreira & Silva, 2020).

What might be a little surprising, however, is that the early contributions of mathematics to consciousness science were few, isolated, and often carried out by non-mathematicians. These issues took a large toll on the quality of mathematical work in the field.

A first trend that we would like to identify concerns the resolution of these issues in recent years in the form of the emergence of a community of researchers who work on mathematical topics. This community is in the process of self-organizing, aiming for a shared sense of purpose within the larger scheme of consciousness science, and seeks to encourage and incentivise open-minded, unprejudiced, and non-dogmatic research. It is in the process of developing checks and balances that ensure that mathematical research in consciousness science lives up to highest scientific and mathematical standards.¹⁹

One important example of the development of checks and balances concerns the usefulness of mathematics. It is important to note that not every mathematical method is necessarily a good method. If applied without proper care, they can have negative consequences for the development of a research programme or field. There is a high risk of over-formalization that can halt progress; unwarranted rigour can slow things down; mathematics can create a barrier in understanding, simply because it is sometimes hard to grasp. If no mathematically-trained peers are available for review, there is even a risk of using mathematics as deception, to hide bad ideas behind walls of formulae. All of these issues have successfully been resolved within mathematics in other disciplines, but resolving them requires a community whose members guard each other against the intended or unintended instantiation of such issues.

It is ensuring to see that the community of mathematicians in consciousness science is forming tools and processes to resolve these issues. Discussion-based improvement is visible in real time. We experience this, for example, in the continuing improvement of the quality of abstracts submitted to conferences and workshops dedicated to mathematical research. A next step to tackle are review-processes that make more systematic use of mathematical expertise, in particular in journals in the field.

¹⁹ The support of this process is the main function of the Association for Mathematical Consciousness Science, which has been founded in 2021 “[t]o connect researchers whose work relates to [Mathematical Consciousness Science (MCS)]”, “to facilitate joint work, exchange of ideas and debates among members in a cooperative and respectful, rather than competitive, atmosphere”, to “encourage and support open-minded, unprejudiced, and non-dogmatic research on MCS while ensuring and advocating scientific quality, honesty and integrity”, and “[t]o offer guidance to the emerging field of MCS through coordinating a discussion of a shared sense of its purpose and its long-term goals” (Association for Mathematical Consciousness Science, 2021).

4.2 Shift from Addition to Improvement

A second trend that we observe at the present concerns the aim of mathematical contributions to the field.

In the past 25 years, mathematics has mainly been used to extend consciousness science. That is, researchers have used mathematics to add new proposals to the wealth of knowledge regarding consciousness.²⁰ This is true of researchers that work within the field, but is particularly noteworthy with respect to researchers who hail from formal disciplines like mathematics or physics. These researchers mainly use mathematics to propose new models, new theories, new hypotheses, or new ideas. But they rarely use mathematics to improve existing models or methods in consciousness science.

Mathematics is a powerful tool, and it makes good sense to use it in formulating new proposals, in particular if these proposals cannot easily be described in ordinary language. But mathematics can also help improve existing research. This is easy to see in the theoretical domain, where theories or models are concerned. But it might also be true for the empirical domain of consciousness science. Experiments make use of statistics, after all, and measurement is usually concerned with numbers (Tal, 2020).

An emerging trend among mathematical approaches in the field is the pursuit of research that is intended to support non-mathematical research projects in consciousness science. Examples are research on the foundations of quality spaces (Kleiner & Ludwig, 2024b), on formal models of emergence (Barnett & Seth, 2023; Hoel et al., 2013; Rosas et al., 2020), on formal representations of pre-reflexive self-awareness (Williford, 2019), on new statistical tools to search for NCCs (Fink et al., 2021; Kleiner, 2025a), and on resolving misunderstandings in the field (Kleiner, 2020), to name just a few.

The vision of a mathematical part of consciousness science that takes the role that mathematical psychology takes in psychology, or mathematical physics takes in physics,—a mathematical consciousness science in the full sense of this term—has arrived in the community, and various research projects are launched in this spirit. It contributes to research on theories of consciousness, on modelling experiments, on conceptual and methodological questions, and on artificial consciousness (see Ch. 5.1 of this volume (“Mathematical

²⁰ The works cited in Chapter 5.1 are notable examples.

Approaches in the Scientific Study of Consciousness”) for details)., . Such a mathematical part of consciousness science should include the pursuit of novel ideas in mathematical terms. But it also needs to live up to the responsibility that the many mathematical problems, questions, and opportunities in consciousness science create.²¹

4.3 Shift from Imported Mathematics to Consciousness-Crafted Math

A third trend, finally, concerns the content of the mathematics that are developed in the field. While previous research in consciousness science has mainly imported mathematics from other scientific fields, there are signs that consciousness science might foster the development of new mathematics, apt to the specific challenges that the phenomenon “consciousness” raises.

Consciousness is a difficult phenomenon to study. Many of the presumptions that other fields rely on cannot be applied to consciousness “just like that”. There are conceptual hurdles that need to be overcome, and differences in epistemological and ontological matters abound. These differences have deep consequences for scientific methodology, including methodology that relies on mathematical tools.

Consider, for example, the mathematical representation of a phenomenon in a scientific theory. While other disciplines usually need not worry too much about how a phenomenon can be represented—phenomena are usually objectively accessible, in which case there is a suite of options for scientific modelling (Frigg & Hartmann, 2025), the question of how to represent consciousness in a scientific theory is far from settled.²² There are no consciousness-meters, and tools to measure consciousness, perhaps in the form of C-Tests (Bayne et al.,

²¹ It is important to note that there is a crucial difference between mathematical and formal research. Whereas the former is concerned with intrinsically mathematical questions—questions that pertain to the quality, logic, rigour, coherence, well-definedness, etc. of a mathematical object or statement—the latter is concerned with the application of mathematical tools and ideas. While crucial, this difference is notoriously difficult to explain in a non-mathematical context. It is perhaps best to think of this difference as analogous to a difference between somebody who builds a measurement device (e.g. an fMRI scanner), and someone who uses that device. Both activities require highly trained researchers, but with entirely different backgrounds and expertise. Similarly, the creation, assessment, and improvement of mathematics require researchers with an entirely different set of backgrounds and expertise from those who make use of the mathematics. Because mathematics is central to many theories and methods, failure to recognize this difference can be a key driver of degenerative research programmes, even in a largely empirical field.

²² A field whose epistemological context is somewhat similar to consciousness science is psychology. But even here, there are noteworthy differences. For example, theorizing plays an entirely different role in psychology than in consciousness science, and measurement theories in psychology, for example the famous representational measurement theory (Krantz et al., 1971), can be much more operational than the phenomenon of consciousness may afford.

2024, Negro & Mudrik, this volume (Ch. 1.7 “Testing for consciousness beyond consensus cases”)) or measures of consciousness (A. K. Seth et al., 2006), or in terms of introspection (Spener, 2024) require the use of intricate logical relationships, and arguably, the development of new conceptual and mathematical tools.

While it is too early to call the shots, we would like to provide three speculative examples of formal research programmes that have the potential to improve science beyond the boundaries of consciousness. First, the qualitative analysis proposed in micro-phenomenology (Petitmengin et al., 2019) may revolutionize qualitative analysis in other fields.²³ Second, the emerging work on a measurement theory for consciousness science (Kleiner, 2025b; Maniscalco et al., 2020; Michel, 2023; Morales, 2021) may spur the “revolution that never happened” in Psychology (Cliff, 1992). Third, very speculatively, the development of more suitable logical and statistical tools to measure neural correlates over contrastive analysis may offer new perspectives and more powerful statistics for measuring the neural correlates of other cognitive or psychological functions or attributes (Chalmers, 2000; Fink, 2020; Kleiner, 2025a; Maschke et al., 2024; Paßler, 2023).

Because of the notorious nature of the challenges that consciousness science faces, if successful, it may well become one of the leading disciplines of the 21st century, much like physics was in the 20th century.

In light of this, young researchers who have a passion for intricate mathematical problems and their relation to reality, are cordially invited to join this exciting new mathematical part of consciousness science, cf. the chapters “Integrated Information Theory” (Ch. 2.5) and “The Qualia Structure Paradigm” (Ch. 5.4) for places to start.

4.4 Towards a New Paradigm for Consciousness Science

In the previous three sections, we have explained how changes in mathematical research programmes in consciousness science, and in the mathematical research community more generally, may contribute to progress or breakthroughs in consciousness science in the upcoming years. In this final section, we would like to follow the editors’ invitation to go a

²³ Micro-phenomenology aims to assess the structure of the experience, and to do so applies a methodology that was derived in spirit from the Semantic Network method (Petitmengin et al., 2019). Hence it may well be viewed as part of the exciting new formal research directions in consciousness science. – Which is not to say that it is also exciting for its non-formal ideas, of course.

step further and provide a speculative case-study of how mathematics might make a difference to the field; a difference in the very paradigm that is applied in consciousness science.

In what follows, we use the term “paradigm” in a roughly Kuhnian sense (Kuhn, 1962): we take the term to refer to the theoretical commitments, methodological norms, types of explanation, and exemplary experiments that shape science within a research community.

Consciousness science is rich in paradigms so understood. There are paradigms of how to build theories of consciousness; paradigms of how to discover the neural or computational substrates of consciousness; paradigms of how to test theories of consciousness; paradigms of how to conceptualize consciousness based on the various problems of consciousness that shape our mind-sets as researchers; and several more.²⁴

Some of these paradigms are already changing. For example, the imminent structural turn in consciousness science is already changing how theories of consciousness are (Kleiner, 2024), how empirical searches for NCCs are conceived (Fink et al., 2021; Kleiner, 2025a), and what may be considered a measurement of consciousness in the first place. The minimal phenomenal experience and non-ordinal states of consciousness research programmes (Facco et al., 2021; Ludwig, 1966; Metzinger, 2020) already make a whole new range of experiences available to consciousness studies; these experiences will likely revolutionize our empirical search for the correlates of consciousness, and can serve as entirely new starting points for the constructions of models and theories of consciousness, as exemplified by the computational phenomenology research programme (Harlan, 1984; Ramstead et al., 2022). In all of these paradigm changes, mathematics plays a crucial role: it can function as a tool to represent conscious experiences, to link consciousness to neuronal models or statistical methods, and,

²⁴ In a nutshell, an essential part of the paradigm on how to build theories of consciousness is what we criticised in section 1. It is centered around associating a binary representation of some aspect of conscious experience—often simply whether a stimulus or signal is consciously experienced—with a binary distinction pertaining to the brain. Another essential part of this paradigm is the disregard of questions regarding measurement of consciousness in the theory-building process. The paradigm of how to discover the neural or computational substrates of consciousness is centered around contrastive analysis (cf. section 1.2), disregarding the fact that contrastive analysis cannot discover the NCC as canonically defined (cf. Kleiner, 2025a). The paradigm on how to test theories of consciousness is based on a particular view on inference and prediction of conscious experiences, which is deeply problematic (cf. Kleiner & Hoel, 2021). And an essential part of the paradigm that shapes our conceptualisation and mind-set as researchers are binary conceptions of consciousness, a consciousness-as-individuable-property view, and intuitions that derive from a particular class of thought-experiments, e.g. philosophical zombies.

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

crucially, to communicate and unify ideas across scientific perspectives and cultural and disciplinary backgrounds.

There is, however, also a larger paradigm at play in consciousness science, that some consider in urgent need of revision. That is the paradigm of objective science itself.

Science, according to Bitbol (2012), Frank et al. (2024), Varela (1996) and others, was formed as inherently “objective”: its development over the last centuries was based on—and perhaps required—a systematic exclusion of subjective conscious experience. But now that science has far advanced and brought subjective experience back as its object of inquiry, it is bound by its past, to treat experience in the same way it treats its other objects of investigation: as a phenomenon that can be modelled, measured, and explained in the same way as other objective phenomena can be modelled, measured, and explained.

The unreflected application of the objective methods of science to subjective experience, according to this line of reasoning, is not simply bound to fail, but may result in a misguided science of consciousness, a sort of un-ending exploration that defeats real progress.

To resolve this problem, then, Consciousness Science must take true subjective experience into account. It needs to develop means to build theories, models, and measures that do not rely on a conceptualization of consciousness as a property akin to the many other objective properties that science discusses, but that integrate subjective experience. This is not a call for a science that is based on the scientist’s own subjective experience alone; rather, it is a call for a science whose tools and methods live up to lived experience, not an abstraction thereof that is made to fit into whichever methods objective science has successfully applied before.

Perhaps surprisingly, it might be mathematics that opens the door for a science of consciousness of this kind.

That is the case because mathematics, understood as a language and mode of thinking, rather than a crude machinery for calculation, offers novel ways of expressing ideas, thoughts,

and phenomena; ways that go beyond what our terminology in the contemporary science of consciousness can offer.²⁵

Mathematics, for example, might be precisely what is needed to make the rich insights on the dynamics, structures, and nature of conscious experiences that are available in religious and contemplative traditions available to science and scientific understanding. Because its expressions are less culturally appropriated, mathematics might be a novel opportunity for a true suspension of conceptual and ontological commitments regarding consciousness, as required by the phenomenological method of epoché (“bracketing”) and analogous notions in contemplative practices. And because of its richness of expressions and constructions, mathematics might, perhaps, afford for a faithful expression of the nuanced experiences reached in contemplative practices.

The vision we describe here is that of a mathematics that can help transcend the limitations of our culturally and scientifically conditioned conceptualisations of consciousness, in the very same way in which it helped transcend the limitations of our evolutionarily conditioned conceptualisations of reality in modern physics, where space, time and matter are concerned.

Clearly, the advent of this vision is still far out of sight, and it might never actually see real light. Perhaps the problem is misguided. We add these considerations here, in response to the kind invitation of the editors, to provide a very tentative sketch of the extent to which mathematical approaches might, if things go well, contribute to change the foundations of the field.²⁶

²⁵ This, of course, depends a lot on what exactly one takes mathematics—and mathematical practice—to be. Edmund Husserl, for example, saw the use of mathematics for abstraction and objectification as one of the main contributors to what he called the crisis in science (Husserl et al., 1982). Husserl’s background was in the calculus of variation (Kockelmans, 1967), a part of mathematics that in that time was intricately connected to the study of partial differential equations, which was bound to the very objective and reifying view of nature of 19th century physics. More modern developments in mathematics, however, and especially the role of mathematics in 20th century physics, deliver an opposing view, where mathematics is precisely what can break the objectification and reification in science. In fact, a more modern view of mathematics implies that “Husserlian doctrines which seem to speak against application of mathematics to phenomenology do not in fact do so” (Yoshimi, 2007).

²⁶ None of the other applications of mathematics we have discussed in earlier sections hinge on this possibility.

Conclusion

Understanding consciousness is one of the most profound scientific challenges of our time. This chapter set out to identify and suggest emerging trends, promising strategies and perspectives in the science of consciousness from the viewpoint of early-career researchers. Besides general, methodological suggestions, we also offered local ones for each section. Our choice of topics is deliberate: we highlight those areas where we believe breakthroughs are most needed, most likely and achievable, and where developments will crucially shape the trajectory of consciousness research in its entirety. These four areas are: the empirical research of human consciousness, theory of consciousness, consciousness in non-human systems and mathematics as an instance of the necessary interdisciplinary nature of the field.

We envision an essentially integrative science of consciousness:

First, in empirical research, it will pursue specificity through structural constraints. At the same time, it will embrace context-recognition to study consciousness as it unfolds in embodied, ecological settings via new technology and large scale ecological experiments.

Second, in the theoretical realm, it will turn from competition to synthesis, recognizing that different theories may illuminate different explananda²⁷ and that real progress lies in synthesizing them for complementary insight rather than forcing premature arbitration.

Third, in extending beyond humans, it will adopt modest biological naturalism for non-human animals, focusing on behavioural markers. For artificial systems, it will adopt a critical theory-guided pluralism and will explore consciousness in basal biological systems, thereby transforming our concepts and ensuring an inclusive and ethically attuned science.

Fourth, in an openness to fundamental restructuring of concepts and methods, it develops its essential interdisciplinarity. As part of this, it will cultivate mathematics and information-theory not only as borrowed tools, but as consciousness-crafted formalisms, building a new community, refining existing models, and exporting innovations to other sciences. Such integration will lead to new paradigms, terminology and concepts.

²⁷ Usually different aspects of „consciousness“.

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

These trajectories point toward the new generation of a field: open, interdisciplinary, and integrative, embracing complementarity rather than rivalry. This vision could embrace consciousness science as one of the defining intellectual projects of the 21st century.

What could the far future of a consciousness science transformed by these changes look like? Here we leave the space of suggestions, predictions and trends and invite the reader to use their imagination.

In the long term, we expect a fundamental restructuring of concepts and terminology. Research branches focusing on various non-human systems and different theories and disciplines each have their own contributions, systems of terminology and concepts. Building bridges is challenging because it requires true openness and willingness to expand and revise existing concepts to be able to find common ground, or at least be able to productively communicate. However, this is exactly why it is so valuable: It forms truly new understanding. While specification of terminology and sharpening of ideas is helpful, artificial competition may waste energy on the extinction of ideas. Emphasizing collaborative integration of perspectives and knowledge synthesization to bridge theories and disciplines, requires effort. Yet, this effort likely results in the gain of revisited, refined, and even novel terminology and concepts.

The concept of “consciousness” and related terms, such as “experience” or “awareness”, are not (and should not be) fixed. It is possible that “consciousness” may, in the future, serve as an umbrella term for a group of revised and new concepts. Just as the search for an *élan vital* became obsolete once the concept of “life” dissolved into the more informative frameworks of biochemistry and molecular biology, so too may today’s problems of consciousness be reshaped as the science itself transforms (Seth, 2021). What could it look like for the puzzle of consciousness to be reframed into several new questions? If our current methods were ill-equipped and we were creating paradoxes by virtue of how we ask our questions, then fundamentally transformed concepts could lead to a new understanding.²⁸ Opening our terminology, theories and concepts to integration of new disciplines and perspectives can show us what such a new science of consciousness looks like. In the long

²⁸ Credit where credit is due: Daniel Dennett and others have long discussed how reframing the “Hard Problem” into multiple more modest questions (e.g., about reportability, illusion, discrimination, access, etc.) removes much of its supposed intractability (Dennett, 1991). Adam Frank, Marcelo Gleiser, and Evan Thompson, in *The Blind Spot* (2019), build on phenomenological traditions to show how modern science established its method by systematically excluding subjectivity and as a consequence now struggles to reintroduce it with the very frameworks that once abstracted it away.

term, the science of consciousness will not simply answer old questions—it will change what it means to ask them.

References

- Andrews, K. (2024). “All animals are conscious”: Shifting the null hypothesis in consciousness science. *Mind & Language*, 39(3), 415–433.
<https://doi.org/10.1111/mila.12498>
- Aru, J., Bachmann, T., Singer, W., & Melloni, L. (2012). Distilling the neural correlates of consciousness. *Neuroscience & Biobehavioral Reviews*, 36(2), Article 2.
<https://doi.org/10.1016/j.neubiorev.2011.12.003>
- Association for Mathematical Consciousness Science. (2021). <https://amcs-community.org/>
- Avneon, M., & Lamy, D. (2018). Reexamining unconscious response priming: A liminal-prime paradigm. *Consciousness and Cognition*, 59, 87–103.
<https://doi.org/10.1016/j.concog.2017.12.006>
- Baars, B. J. (1986). What is a Theory of Consciousness a Theory Of? The Search for Criterial Constraints on Theory. *Imagination, Cognition, and Personality*, 1, 3–24.
- Baciadonna, L., Macri, C., & Giurfa, M. (2023). Associative learning: Unmet criterion for plant sentience. *Animal Sentience*, 8(33). <https://doi.org/10.51291/2377-7478.1809>
- Baluška, F., Miller, W. B., Slijepcevic, P., & Reber, A. S. (2025). Sensing, feeling and sentience in unicellular organisms and living cells. *BioSystems*, 247, 105374.
<https://doi.org/10.1016/j.biosystems.2024.105374>
- Baluška, F., & Reber, A. (2019). Sentience and Consciousness in Single Cells: How the First Minds Emerged in Unicellular Species. *BioEssays*, 41(3).
<https://doi.org/10.1002/bies.201800229>
- Barandiaran, X., & Moreno, A. (2006). On What Makes Certain Dynamical Systems Cognitive: A Minimally Cognitive Organization Program. *Adaptive Behavior*, 14(2), 171–185. <https://doi.org/10.1177/105971230601400208>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Barnett, L., & Seth, A. K. (2023). Dynamical independence: Discovering emergent macroscopic processes in complex dynamical systems. *Physical Review E*, 108(1), 014304. <https://doi.org/10.1103/PhysRevE.108.014304>

Barron, A. B., & Klein, C. (2016). What insects can tell us about the origins of consciousness. *Proceedings of the National Academy of Sciences*, 113(18), 4900–4908. <https://doi.org/10.1073/pnas.1520084113>

Bayne, T., Seth, A. K., Massimini, M., Shepherd, J., Cleeremans, A., Fleming, S. M., Malach, R., Mattingley, J. B., Menon, D. K., Owen, A. M., Peters, M. A. K., Razi, A., & Mudrik, L. (2024). Tests for consciousness in humans and beyond. *Trends in Cognitive Sciences*, 28(5), 454–466. <https://doi.org/10.1016/j.tics.2024.01.010>

Biafora, M., & Schmidt, T. (2020). Induced dissociations: Opposite time courses of priming and masking induced by custom-made mask-contrast functions. *Attention, Perception, & Psychophysics*, 82(3), 1333–1354. <https://doi.org/10.3758/s13414-019-01822-4>

Birch, J. (2022). The search for invertebrate consciousness. *Noûs*, 56(1), 133–153. <https://doi.org/10.1111/nous.12351>

Birch, J., Ginsburg, S., & Jablonka, E. (2020). Unlimited Associative Learning and the origins of consciousness: A primer and some predictions. *Biology & Philosophy*, 35(6). <https://doi.org/10.1007/s10539-020-09772-0>

Bitbol, M. (2012). Neurophenomenology, an Ongoing Practice of/in Consciousness. *Constructivist Foundations*, 7(3), 165–173.

Block, N. (2011). Perceptual consciousness overflows cognitive access. *Trends in Cognitive Sciences*, 15(12), 567–575. <https://doi.org/10.1016/j.tics.2011.11.001>

Block, N. (2019). What Is Wrong with the No-Report Paradigm and How to Fix It. *Trends in Cognitive Sciences*, 23(12), 1003–1013. <https://doi.org/10.1016/j.tics.2019.10.001>

Blum, M., & Blum, L. (2020). A Theoretical Computer Science Perspective on Consciousness (Version 4). *arXiv*. <https://doi.org/10.48550/ARXIV.2011.09850>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Boisseau, R. P., Vogel, D., & Dussutour, A. (2016). Habituation in non-neural organisms: Evidence from slime moulds. *Proceedings of the Royal Society B: Biological Sciences*, 283(1829), 20160446. <https://doi.org/10.1098/rspb.2016.0446>

Bronfman, Z. Z., Ginsburg, S., & Jablonka, E. (2016). The Transition to Minimal Consciousness through the Evolution of Associative Learning. *Frontiers in Psychology*, 7. <https://doi.org/10.3389/fpsyg.2016.01954>

Butlin, P., Long, R., Elmoznino, E., Bengio, Y., Birch, J., Constant, A., Deane, G., Fleming, S. M., Frith, C., Ji, X., Kanai, R., Klein, C., Lindsay, G., Michel, M., Mudrik, L., Peters, M. A. K., Schwitzgebel, E., Simon, J., & VanRullen, R. (2023). Consciousness in Artificial Intelligence: Insights from the Science of Consciousness (Version 3). *arXiv*. <https://doi.org/10.48550/ARXIV.2308.08708>

Cabral-Calderin, Y., Hechavarria, J., & Melloni, L. (2025). Towards a neuroethological approach to consciousness. *Center for Open Science*. https://doi.org/10.31234/osf.io/wyrhu_v2

Calvo Garzón, P., & Keijzer, F. (2011). Plants: Adaptive behavior, root-brains, and minimal cognition. *Adaptive Behavior*, 19(3), 155–171. <https://doi.org/10.1177/1059712311409446>

Cao, R. (2022). Multiple realizability and the spirit of functionalism. *Synthese*, 200(6). <https://doi.org/10.1007/s11229-022-03524-1>

Carruthers, P. (2019). *Human and Animal Minds: The Consciousness Questions Laid to Rest* (1st ed.). Oxford University Press. <https://doi.org/10.1093/oso/9780198843702.001.0001>

Cartwright, N. (1983). *How the Laws of Physics Lie* (1st ed.). Oxford University PressOxford. <https://doi.org/10.1093/0198247044.001.0001>

Casali, A. G., Gosseries, O., Rosanova, M., Boly, M., Sarasso, S., Casali, K. R., Casarotto, S., Bruno, M.-A., Laureys, S., Tononi, G., & Massimini, M. (2013). A Theoretically Based Index of Consciousness Independent of Sensory Processing and Behavior. *Science Translational Medicine*, 5(198), Article 198. <https://doi.org/10.1126/scitranslmed.3006294>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Chalmers, D. J. (1996). *The conscious mind: In search of a fundamental theory*. Oxford University Press.

Chalmers, D. J. (2000). What is a neural correlate of consciousness? In *Neural correlates of consciousness: Empirical and conceptual questions* (pp. 17–39). The MIT Press.

Chen, H., & Wyble, B. (2015). Amnesia for Object Attributes: Failure to Report Attended Information That Had Just Reached Conscious Awareness. *Psychological Science*, 26(2), Article 2. <https://doi.org/10.1177/0956797614560648>

Chittka, L. (2017). Bee cognition. *Current Biology*, 27(19), R1049–R1053. <https://doi.org/10.1016/j.cub.2017.08.008>

Clark, A. (1993). *Sensory Qualities*. Oxford University Press UK.

Cliff, N. (1992). Article Commentary: Abstract Measurement Theory and the Revolution that Never Happened. *Psychological Science*, 3(3), 186–190. <https://doi.org/10.1111/j.1467-9280.1992.tb00024.x>

Cohen, M. A., Botch, T. L., & Robertson, C. E. (2020). The limits of color awareness during active, real-world vision. *Proceedings of the National Academy of Sciences*, 117(24), 13821–13827. <https://doi.org/10.1073/pnas.1922294117>

Cohen, M. A., & Dennett, D. C. (2011). Consciousness cannot be separated from function. *Trends in Cognitive Sciences*, 15(8), 358–364. <https://doi.org/10.1016/j.tics.2011.06.008>

Cohen, M. A., Dennett, D. C., & Kanwisher, N. (2016). What is the Bandwidth of Perceptual Experience? *Trends in Cognitive Sciences*, 20(5), 324–335. <https://doi.org/10.1016/j.tics.2016.03.006>

Cohen, M. A., Konkle, T., Rhee, J. Y., Nakayama, K., & Alvarez, G. A. (2014). Processing multiple visual objects is limited by overlap in neural channels. *Proceedings of the National Academy of Sciences*, 111(24), 8955–8960. <https://doi.org/10.1073/pnas.1317860111>

Cohen, M. A., Nakayama, K., Konkle, T., Stantić, M., & Alvarez, G. A. (2015). Visual Awareness Is Limited by the Representational Architecture of the Visual System. *Journal of Cognitive Neuroscience*, 27(11), Article 11. https://doi.org/10.1162/jocn_a_00855

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Corcoran, A. W., Haun, A. M., Dorman, R., Tononi, G., Friston, K. J., Pennartz, C. M. A., TWCF, :, & Consortium, I. (2025). Integrated information and predictive processing theories of consciousness: An adversarial collaborative review (Version 1). arXiv. <https://doi.org/10.48550/ARXIV.2509.00555>

Crick, F., & Koch, C. (1990). Towards a neurobiological theory of consciousness. *Seminars in the Neurosciences*, 2(263–275), 203. <https://digirepo.nlm.nih.gov/ext/document/101584582X469/PDF/101584582X469.pdf>

Crump, A., Browning, H., Schnell, A., Burn, C., & Birch, J. (2022). Sentience in decapod crustaceans: A general framework and review of the evidence. *Animal Sentience*, 7(32). <https://doi.org/10.51291/2377-7478.1691>

de Gardelle, V., Charles, L., & Kouider, S. (2011). Perceptual awareness and categorical representation of faces: Evidence from masked priming. *Consciousness and Cognition*, 20(4), Article 4. <https://doi.org/10.1016/j.concog.2011.02.001>

Dehaene, S., Kerszberg, M., & Changeux, J.-P. (1998). A Neuronal Model of a Global Workspace in Effortful Cognitive Tasks. *Proceedings of the National Academy of Sciences of the United States of America*, 95(24), Article 24.

Dehaene, S., Naccache, L., Clec'H, G. L., Koechlin, E., Mueller, M., & Dehaene-Lambertz, G. (1998). Imaging unconscious semantic priming. 395, 4.

Dehaene, S., Naccache, L., Cohen, L., Bihan, D. L., Mangin, J.-F., Poline, J.-B., & Rivière, D. (2001). Cerebral mechanisms of word masking and unconscious repetition priming. *Nature Neuroscience*, 4(7), Article 7. <https://doi.org/10.1038/89551>

Dembski, C., Koch, C., & Pitts, M. (2021). Perceptual awareness negativity: A physiological correlate of sensory consciousness. *Trends in Cognitive Sciences*, 25(8), 660–670. <https://doi.org/10.1016/j.tics.2021.05.009>

Dennett, D. C. (1991). *Consciousness explained* (pp. xiii, 511). Little, Brown and Co.

Down, K. J. A., & Mediano, P. A. M. (2023). A Logarithmic Decomposition for Information (Version 1). arXiv. <https://doi.org/10.48550/ARXIV.2305.07554>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Duncan, I. J. H. (2006). The changing concept of animal sentience. *Applied Animal Behaviour Science*, 100(1), 11–19. <https://doi.org/10.1016/j.applanim.2006.04.011>

Dung, L. (2023). How to Deal with Risks of Ai Suffering. *Inquiry: An Interdisciplinary Journal of Philosophy*. <https://doi.org/10.1080/0020174x.2023.2238287>

Dung, L. (2025). Tests of Animal Consciousness are Tests of Machine Consciousness. *Erkenntnis*, 90(4), 1323–1342. <https://doi.org/10.1007/s10670-023-00753-9>

Dussutour, A. (2021). Learning in single cell organisms. *Biochemical and Biophysical Research Communications*, 564, 92–102. <https://doi.org/10.1016/j.bbrc.2021.02.018>

Dziri, N., Lu, X., Sclar, M., Li, X. L., Jiang, L., Lin, B. Y., West, P., Bhagavatula, C., Bras, R. L., Hwang, J. D., Sanyal, S., Welleck, S., Ren, X., Ettinger, A., Harchaoui, Z., & Choi, Y. (2023). Faith and Fate: Limits of Transformers on Compositionality (Version 3). *arXiv*. <https://doi.org/10.48550/ARXIV.2305.18654>

Endersby, J. (2009). Lumpers and Splitters: Darwin, Hooker, and the Search for Order. *Science*, 326(5959), 1496–1499. <https://doi.org/10.1126/science.1165915>

Facco, E., Fracas, F., & Tressoldi, P. (2021). Moving beyond the concept of altered state of consciousness: The Non-Ordinary Mental Expressions (NOMEs). *Advances in Social Sciences Research Journal*, 8(3), 615–631. <https://doi.org/10.14738/assrj.83.9935>

Fahrenfort, J. J., Johnson, P. A., Kloosterman, N. A., Stein, T., & van Gaal, S. (2025). Criterion placement threatens the construct validity of neural measures of consciousness. *eLife*, 13, RP102335. <https://doi.org/10.7554/eLife.102335>

Fazekas, P., Cleeremans, A., & Overgaard, M. (2024). A construct-first approach to consciousness science. *Neuroscience & Biobehavioral Reviews*, 156, 105480. <https://doi.org/10.1016/j.neubiorev.2023.105480>

Ferrante, O., Gorska-Klimowska, U., Henin, S., Hirschhorn, R., Khalaf, A., Lepauvre, A., Liu, L., Richter, D., Vidal, Y., Bonacchi, N., Brown, T., Sripad, P., Armendariz, M., Bendtz, K., Ghafari, T., Hetenyi, D., Jeschke, J., Kozma, C., Mazumder, D. R., ... Melloni, L. (2025). Adversarial testing of global neuronal workspace and integrated information theories of consciousness. *Nature*, 642(8066), 133–142. <https://doi.org/10.1038/s41586-025-08888-1>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Ferreira, C., & Silva, C. (2020). The Roles of Mathematics in the History of Science: The Mathematization Thesis. *Transversal: International Journal for the Historiography of Science*, 8. <https://doi.org/10.24117/2526-2270.2020.i8.03>

Fink, S. B. (2020). A double anniversary for the neural correlates of consciousness: Editorial introduction. *Philosophy and the Mind Sciences*, 1(II). <https://doi.org/10.33735/phimisci.2020.ii.85>

Fink, S. B., Kob, L., & Lyre, H. (2021). A structural constraint on neural correlates of consciousness. *Philosophy and the Mind Sciences*, 2. <https://doi.org/10.33735/phimisci.2021.79>

Francken, J. C., Beerendonk, L., Molenaar, D., Fahrenfort, J. J., Kiverstein, J. D., Seth, A. K., & Van Gaal, S. (2022). An academic survey on theoretical foundations, common assumptions and the current state of consciousness science. *Neuroscience of Consciousness*, 2022(1). <https://doi.org/10.1093/nc/niac011>

Frank, A., Gleiser, M., & Thompson, E. (2024). *The blind spot: Why science cannot ignore human experience*. The MIT Press.

Freud, S. (1923). The Ego and the Id and Other Works. In J. Strachey (Ed. & Trans.), *The Standard Edition of the Complete Psychological Works of Sigmund Freud: Vol. XIX (1923-1925)* (pp. 1–308). London: Hogarth Press.

Frigg, R., & Hartmann, S. (2025). Models in Science. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford Encyclopedia of Philosophy* (Summer 2025). Metaphysics Research Lab, Stanford University. <https://plato.stanford.edu/archives/sum2025/entries/models-science/>

Fu, Y., Yan, W., Shen, M., & Chen, H. (2021). Does consciousness overflow cognitive access? Novel insights from the new phenomenon of attribute amnesia. *Science China Life Sciences*, 64(6), 847–860. <https://doi.org/10.1007/s11427-020-1831-8>

Gallup, G. G. (1970). Chimpanzees: Self-Recognition. *Science*, 167(3914), 86–87. <https://doi.org/10.1126/science.167.3914.86>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Gershman, S. J., Balbi, P. E., Gallistel, C. R., & Gunawardena, J. (2021). Reconsidering the evidence for learning in single cells. *eLife*, 10. <https://doi.org/10.7554/elife.61907>

Godfrey-Smith, P. (2016a). Individuality, subjectivity, and minimal cognition. *Biology & Philosophy*, 31(6), 775–796. <https://doi.org/10.1007/s10539-016-9543-1>

Godfrey-Smith, P. (2016b). *Other minds: The octopus, the sea, and the deep origins of consciousness* (First edition). Farrar, Straus and Giroux.

Godfrey-Smith, P. (2019). Evolving Across the Explanatory Gap. *Philosophy, Theory, and Practice in Biology*, 11(20220112). <https://doi.org/10.3998/ptpbio.16039257.0011.001>

Godfrey-Smith, P. (2021). Learning and the biology of consciousness: A commentary on Birch, Ginsburg, and Jablonka. *Biology & Philosophy*, 36(5), 44. <https://doi.org/10.1007/s10539-021-09820-3>

Griffin, D. R. (1977). Anthropomorphism. *BioScience*, 27(7), 445–446. <https://doi.org/10.2307/1297524>

Griffin, D. R. (1981). *The Question of Animal Awareness: Evolutionary Continuity of Mental Experience*. Rockefeller Univ. Press.

Gusev, E., Sarapultsev, A., & Komelkova, M. (2025). Evolutionary Trajectories of Consciousness: From Biological Foundations to Technological Horizons. *Brain Sciences*, 15(7), 734. <https://doi.org/10.3390/brainsci15070734>

Haberkamp, A., Schmidt, F., Biafora, M., & Schmidt, T. (2019). Interpreting and responding to ambiguous natural images in spider phobia. *Journal of Behavior Therapy and Experimental Psychiatry*, 65, 101495. <https://doi.org/10.1016/j.jbtep.2019.101495>

Haken, H., & Haken, H. (1996). *Principles of Brain Functioning* (Vol. 67). Springer Berlin Heidelberg. <https://doi.org/10.1007/978-3-642-79570-1>

Hansen, M. J. (2024). A critical review of plant sentience: Moving beyond traditional approaches. *Biology & Philosophy*, 39(4). <https://doi.org/10.1007/s10539-024-09953-1>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Harlan, R. M. (1984). Towards a Computational Phenomenology. In J. N. Mohanty (Ed.), *Phenomenology and the Human Sciences* (pp. 21–37). Springer Netherlands.

https://doi.org/10.1007/978-94-009-5081-8_3

Hesselmann, G., Darcy, N., Sterzer, P., & Knops, A. (2015). Exploring the boundary conditions of unconscious numerical priming effects with continuous flash suppression. *Consciousness and Cognition*, 31, 60–72. <https://doi.org/10.1016/j.concog.2014.10.009>

Hirschhorn, R., Biderman, D., Biderman, N., Yaron, I., Bennet, R., Plotnik, M., & Mudrik, L. (2024). Using virtual reality to induce multi-trial inattentional blindness despite trial-by-trial measures of awareness. *Behavior Research Methods*, 56(4), 3452–3468.

<https://doi.org/10.3758/s13428-024-02401-8>

Hoel, E. P., Albantakis, L., & Tononi, G. (2013). Quantifying causal emergence shows that macro can beat micro. *Proceedings of the National Academy of Sciences*, 110(49), 19790–19795. <https://doi.org/10.1073/pnas.1314922110>

Husserl, E., Husserl, E., & Husserl, E. (1982). General introduction to a pure phenomenology. M. Nijhoff ; Distributors for the U.S. and Canada, Kluwer Boston.

Jablonka, E., & Ginsburg, S. (2022). Learning and the Evolution of Conscious Agents. *Biosemiotics*, 15(3), 401–437. <https://doi.org/10.1007/s12304-022-09501-y>

Jékely, G., Godfrey-Smith, P., & Keijzer, F. (2021). Reafference and the origin of the self in early nervous system evolution. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 376(1821), 20190764. <https://doi.org/10.1098/rstb.2019.0764>

Kahneman, D. (2003). Experiences of collaborative research. *American Psychologist*, 58(9), 723–730. <https://doi.org/10.1037/0003-066X.58.9.723>

Katlowitz, K. A., Belanger, J. L., Ismail, T., Chavez, A. G., Chericoni, A., Franch, M., Mickiewicz, E. A., Mathura, R. K., Paulo, D. L., Bartoli, E., Piantadosi, S. T., Provenza, N. R., Watrous, A. J., Sheth, S. A., & Hayden, B. Y. (2025). Attention is all you need (in the brain): Semantic contextualization in human hippocampus. *Neuroscience*.

<https://doi.org/10.1101/2025.06.23.661103>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Kawakita, G., Zeleznikow-Johnston, A., Takeda, K., Tsuchiya, N., & Oizumi, M. (2023). Is my “red” your “red”? Unsupervised alignment of qualia structures via optimal transport [Preprint]. PsyArXiv. <https://doi.org/10.31234/osf.io/h3pqm>

Keijzer, F. (2015a). Moving and sensing without input and output: Early nervous systems and the origins of the animal sensorimotor organization. *Biology & Philosophy*, 30(3), 311–331. <https://doi.org/10.1007/s10539-015-9483-1>

Keijzer, F. (2015b). Moving and sensing without input and output: Early nervous systems and the origins of the animal sensorimotor organization. *Biology & Philosophy*, 30(3), 311–331. <https://doi.org/10.1007/s10539-015-9483-1>

King, J.-R., Pescetelli, N., & Dehaene, S. (2016). Brain Mechanisms Underlying the Brief Maintenance of Seen and Unseen Sensory Information. *Neuron*, 92(5), Article 5. <https://doi.org/10.1016/j.neuron.2016.10.051>

Kleiner, J. (2020). Brain states matter. A reply to the unfolding argument. *Consciousness and Cognition*, 85, 102981. <https://doi.org/10.1016/j.concog.2020.102981>

Kleiner, J. (2024). Towards a structural turn in consciousness science. *Consciousness and Cognition*, 119, 103653. <https://doi.org/10.1016/j.concog.2024.103653>

Kleiner, J. (2025a). On the Logic of Measuring Neural Correlates of Consciousness. PsyArXiv. https://doi.org/10.31234/osf.io/p9ca3_v1

Kleiner, J. (2025b). Topics in mathematical consciousness science [Ludwig-Maximilians-Universität München; Application/pdf]. <https://doi.org/10.5282/EDOC.35420>

Kleiner, J., & Hoel, E. (2021). Falsification and consciousness. *Neuroscience of Consciousness*, 2021(1). <https://doi.org/10.1093/nc/niab001>

Kleiner, J., & Ludwig, T. (2024a). The case for neurons: A no-go theorem for consciousness on a chip. *Neuroscience of Consciousness*, 2024(1). <https://doi.org/10.1093/nc/niae037>

Kleiner, J., & Ludwig, T. (2024b). What is a mathematical structure of conscious experience? *Synthese*, 203(3), 89. <https://doi.org/10.1007/s11229-024-04503-4>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Kockelmans, J. J. (with Internet Archive). (1967). *Phenomenology: The philosophy of Edmund Husserl and its interpretation*. Garden City, N.Y. : Anchor Books.
<http://archive.org/details/phenomenology0000jose>

Koechlin, E., Naccache, L., Block, E., & Dehaene, S. (1999). Primed numbers: Exploring the modularity of numerical representations with masked and unmasked semantic priming. *Journal of Experimental Psychology: Human Perception and Performance*, 25(6), 1882–1905.
<https://doi.org/10.1037/0096-1523.25.6.1882>

Koivisto, M., & Grassini, S. (2018). Unconscious response priming during continuous flash suppression. *PLOS ONE*, 13(2), Article 2. <https://doi.org/10.1371/journal.pone.0192201>

Kouider, S., Gardelle, V. de, Sackur, J., & Dupoux, E. (2010). How rich is consciousness? The partial awareness hypothesis. *Trends in Cognitive Sciences*, 14(7), 301–307.
<https://doi.org/10.1016/j.tics.2010.04.006>

Krantz, D., Luce, D., Suppes, P., & Tversky, A. (Eds.). (1971). *Foundations of Measurement, Vol. I: Additive and Polynomial Representations*. New York Academic Press.

Kuhn, R. L. (2024). A landscape of consciousness: Toward a taxonomy of explanations and implications. *Progress in Biophysics and Molecular Biology*, 190, 28–169.
<https://doi.org/10.1016/j.pbiomolbio.2023.12.003>

Kuhn, T. S. (1962). *The structure of scientific revolutions*. Chicago.

Ladyman, J. (1998). What is structural realism? *Studies in History and Philosophy of Science Part A*, 29(3), 409–424. [https://doi.org/10.1016/S0039-3681\(98\)80129-5](https://doi.org/10.1016/S0039-3681(98)80129-5)

Landman, R., Spekreijse, H., & Lamme, V. A. F. (2003). Large capacity storage of integrated objects before change blindness. *Vision Research*, 43(2), 149–164.
[https://doi.org/10.1016/S0042-6989\(02\)00402-9](https://doi.org/10.1016/S0042-6989(02)00402-9)

Lau, H. (2023). Where is the ‘posterior hot zone’? Open Review of Cogitate et al (2023): “An Adversarial Collaboration to Critically Evaluate Theories of Consciousness.” Center for Open Science. <https://doi.org/10.31234/osf.io/93ufe>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Lepauvre, A. (2025). *Temporal Crossroads*. Radboud University Press.
<https://doi.org/10.54195/9789465150550>

Lepauvre, A., & Melloni, L. (2021). The search for the neural correlate of consciousness: Progress and challenges. *Philosophy and the Mind Sciences*, 2.
<https://doi.org/10.33735/phimisci.2021.87>

Lewis-Healey, E., Tagliazucchi, E., Canales-Johnson, A., & Bekinschtein, T. A. (2024). Breathwork-induced psychedelic experiences modulate neural dynamics. *Cerebral Cortex*, 34(8). <https://doi.org/10.1093/cercor/bhae347>

Lin, Z., & Murray, S. O. (2014). Priming of awareness or how not to measure visual awareness. *Journal of Vision*, 14(1), Article 1. <https://doi.org/10.1167/14.1.27>

Ludwig, A. M. (1966). Altered States of Consciousness. *Archives of General Psychiatry*, 15(3), 225. <https://doi.org/10.1001/archpsyc.1966.01730150001001>

Luppi, A. I., Rosas, F. E., Mediano, P. A. M., Menon, D. K., & Stamatakis, E. A. (2024). Information decomposition and the informational architecture of the brain. *Trends in Cognitive Sciences*, 28(4), 352–368. <https://doi.org/10.1016/j.tics.2023.11.005>

Lutz, A., Lachaux, J.-P., Martinerie, J., & Varela, F. J. (2002). Guiding the study of brain dynamics by using first-person data: Synchrony patterns correlate with ongoing conscious states during a simple visual task. *Proceedings of the National Academy of Sciences*, 99(3), 1586–1591. <https://doi.org/10.1073/pnas.032658199>

Lyon, P. (2006). The biogenic approach to cognition. *Cognitive Processing*, 7(1), 11–29. <https://doi.org/10.1007/s10339-005-0016-8>

Lyon, P., Keijzer, F., Arendt, D., & Levin, M. (2021). Reframing cognition: Getting down to biological basics. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 376(1820), 20190750. <https://doi.org/10.1098/rstb.2019.0750>

Malach, R. (2021). Local neuronal relational structures underlying the contents of human conscious experience. *Neuroscience of Consciousness*, 2021(2), niab028.
<https://doi.org/10.1093/nc/niab028>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Maniscalco, B., Castaneda, O. G., Odegaard, B., Morales, J., Rajananda, S., Denison, R., & Peters, M. A. K. (2020). The relative psychometric function: A general analysis framework for relating psychological processes. Center for Open Science.
<https://doi.org/10.31234/osf.io/5qrjn>

Maniscalco, B., & Lau, H. (2016). The signal processing architecture underlying subjective reports of sensory awareness. *Neuroscience of Consciousness*, 2016(1), niw002.
<https://doi.org/10.1093/nc/niw002>

Maschke, C., O'Byrne, J., Colombo, M. A., Boly, M., Gosseries, O., Laureys, S., Rosanova, M., Jerbi, K., & Blain-Moraes, S. (2024). Critical dynamics in spontaneous EEG predict anesthetic-induced loss of consciousness and perturbational complexity. *Communications Biology*, 7(1). <https://doi.org/10.1038/s42003-024-06613-8>

Mashour, G. A., Roelfsema, P., Changeux, J.-P., & Dehaene, S. (2020). Conscious Processing and the Global Neuronal Workspace Hypothesis. *Neuron*, 105(5), Article 5.
<https://doi.org/10.1016/j.neuron.2020.01.026>

McClelland, T. (2024). Agnosticism About Artificial Consciousness (Version 1). arXiv.
<https://doi.org/10.48550/ARXIV.2412.13145>

McKusick, V. A. (1969). On Lumpers and Splitters, or the Nosology of Genetic Disease. *Perspectives in Biology and Medicine*, 12(2), 298–312.
<https://doi.org/10.1353/pbm.1969.0039>

Melloni, L., Mudrik, L., Pitts, M., Bendtz, K., Ferrante, O., Gorska, U., Hirschhorn, R., Khalaf, A., Kozma, C., Lepauvre, A., Liu, L., Mazumder, D., Richter, D., Zhou, H., Blumenfeld, H., Boly, M., Chalmers, D. J., Devore, S., Fallon, F., ... Tononi, G. (2023). An adversarial collaboration protocol for testing contrasting predictions of global neuronal workspace and integrated information theory. *PLOS ONE*, 18(2), e0268577.
<https://doi.org/10.1371/journal.pone.0268577>

Melloni, L., Mudrik, L., Pitts, M., & Koch, C. (2021). Making the hard problem of consciousness easier. *Science*, 372(6545), Article 6545.
<https://doi.org/10.1126/science.abj3259>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Merleau-Ponty, M. (2013). *Phenomenology of Perception* (0 ed.). Routledge.
<https://doi.org/10.4324/9780203720714>

Metzinger, T. (2020). Minimal phenomenal experience: Meditation, tonic alertness, and the phenomenology of “pure” consciousness. *Philosophy and the Mind Sciences*, 1(I), 1–44.
<https://doi.org/10.33735/phimisci.2020.I.46>

Michel, M. (2023). Calibration in Consciousness Science. *Erkenntnis*, 88(2), 829–850.
<https://doi.org/10.1007/s10670-021-00383-z>

Michel, M., & Lau, H. (2021). Is blindsight possible under signal detection theory? Comment on Phillips (2021). *Psychological Review*, 128(3), 585–591.
<https://doi.org/10.1037/rev0000266>

Micher, N., & Lamy, D. (2023). The role of conscious perception in semantic processing: Testing the action trigger hypothesis. *Consciousness and Cognition*, 107, 103438.
<https://doi.org/10.1016/j.concog.2022.103438>

Mitchell, S. D. (2012). *Unsimple truths: Science, complexity, and policy* (Paperback ed). The Univ. of Chicago Press.

Morales, J. (2021). Introspection Is Signal Detection. *The British Journal for the Philosophy of Science*. <https://doi.org/10.1086/715184>

Mudrik, L., Hirschhorn, R., & Korisky, U. (2024). Taking consciousness for real: Increasing the ecological validity of the study of conscious vs. unconscious processes. *Neuron*, 112(10), 1642–1656. <https://doi.org/10.1016/j.neuron.2024.03.031>

Naccache, L., Blandin, E., & Dehaene, S. (2002). Unconscious Masked Priming Depends on Temporal Attention. *Psychological Science*, 13(5), Article 5. <https://doi.org/10.1111/1467-9280.00474>

Naccache, L., & Dehaene, S. (2001). Unconscious semantic priming extends to novel unseen stimuli. *Cognition*, 80(3), Article 3. [https://doi.org/10.1016/S0010-0277\(00\)00139-6](https://doi.org/10.1016/S0010-0277(00)00139-6)

Naccache, L., Marti, S., Sitt, J. D., Trübutschek, D., & Berkovitch, L. (2016). Why the P3b is still a plausible correlate of conscious access? A commentary on Silverstein et al., 2015. *Cortex*, 85, 126–128. <https://doi.org/10.1016/j.cortex.2016.04.003>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Nagel, T. (1974). What Is It Like to Be a Bat? *The Philosophical Review*, 83(4), 435–450. <https://doi.org/10.2307/2183914>

Neumann, J. von, & Kurzweil, R. (2012). *The Computer and the Brain*. Yale University Press.

Newell, A., Shaw, J. C., & Simon, H. A. (1958). Elements of a theory of human problem solving. *Psychological Review*, 65(3), 151–166. <https://doi.org/10.1037/h0048495>

Newen, A., Gallagher, S., & De Bruin, L. (2018). 4E Cognition: Historical Roots, Key Concepts, and Central Issues. In *The Oxford Handbook of 4E Cognition* (1st ed., pp. 3–16). Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780198735410.013.1>

Northoff, G., & Lamme, V. (2020). Neural signs and mechanisms of consciousness: Is there a potential convergence of theories of consciousness in sight? *Neuroscience & Biobehavioral Reviews*, 118, 568–587. <https://doi.org/10.1016/j.neubiorev.2020.07.019>

Northoff, G., & Zilio, F. (2022). Temporo-spatial Theory of Consciousness (TTC) – Bridging the gap of neuronal activity and phenomenal states. *Behavioural Brain Research*, 424, 113788. <https://doi.org/10.1016/j.bbr.2022.113788>

Overgaard, M. (2021). Insect Consciousness. *Frontiers in Behavioral Neuroscience*, 15, 653041. <https://doi.org/10.3389/fnbeh.2021.653041>

Overgaard, M., Rote, J., Mouridsen, K., & Ramsøy, T. Z. (2006). Is conscious perception gradual or dichotomous? A comparison of report methodologies during a visual task. *Consciousness and Cognition*, 15(4), Article 4. <https://doi.org/10.1016/j.concog.2006.04.002>

Paßler, M. (2023). The exclusionary approach to consciousness. *Neuroscience of Consciousness*, 2023(1). <https://doi.org/10.1093/nc/niad022>

Petitmengin, C., Remillieux, A., & Valenzuela-Moguillansky, C. (2019). Discovering the structures of lived experience: Towards a micro-phenomenological analysis method. *Phenomenology and the Cognitive Sciences*, 18(4), 691–730. <https://doi.org/10.1007/s11097-018-9597-4>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Phillips, I. (2021). Blindsight is qualitatively degraded conscious vision. *Psychological Review*, 128(3), 558–584. <https://doi.org/10.1037/rev0000254>

Pinto, Y., Sligte, I. G., Shapiro, K. L., & Lamme, V. A. F. (2013). Fragile visual short-term memory is an object-based and location-specific store. *Psychonomic Bulletin & Review*, 20(4), 732–739. <https://doi.org/10.3758/s13423-013-0393-4>

Ponte, G., Chiandetti, C., Edelman, D. B., Imperadore, P., Pieroni, E. M., & Fiorito, G. (2022). Cephalopod Behavior: From Neural Plasticity to Consciousness. *Frontiers in Systems Neuroscience*, 15, 787139. <https://doi.org/10.3389/fnsys.2021.787139>

Popper, K. R. (1983). *Realism and the Aim of Science: From the Postscript to the Logic of Scientific Discovery* (W. W. Bartley, Ed.). Routledge.

Premack, D., & Woodruff, G. (1978). Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences*, 1(4), 515–526. <https://doi.org/10.1017/S0140525X00076512>

Ramstead, M. J. D., Seth, A. K., Hesp, C., Sandved-Smith, L., Mago, J., Lifshitz, M., Pagnoni, G., Smith, R., Dumas, G., Lutz, A., Friston, K., & Constant, A. (2022). From Generative Models to Generative Passages: A Computational Approach to (Neuro) Phenomenology. *Review of Philosophy and Psychology*, 13(4), 829–857. <https://doi.org/10.1007/s13164-021-00604-y>

Realization and Multiple Realization. (2016). In T. W. Polger, *The Multiple Realization Book* (pp. 18–36). Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780198732891.003.0002>

Ro, T., Singhal, N. S., Breitmeyer, B. G., & Garcia, J. O. (2009). Unconscious processing of color and form in metacontrast masking. *Perception & Psychophysics*, 71(1), 95–103. <https://doi.org/10.3758/APP.71.1.95>

Rosas, F. E., Mediano, P. A. M., Jensen, H. J., Seth, A. K., Barrett, A. B., Carhart-Harris, R. L., & Bor, D. (2020). Reconciling emergences: An information-theoretic approach to identify causal emergence in multivariate data. *PLOS Computational Biology*, 16(12), e1008289. <https://doi.org/10.1371/journal.pcbi.1008289>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Rothkirch, M., Shanks, D. R., & Hesselmann, G. (2022). The Pervasive Problem of Post Hoc Data Selection in Studies on Unconscious Processing: A Reply to Sklar, Goldstein, and Hassin (2021). *Experimental Psychology*, 69(1), 1–11. <https://doi.org/10.1027/1618-3169/a000541>

Rouleau, N., & Levin, M. (2025). Brains and Where Else? Mapping Theories of Consciousness to Unconventional Embodiments. *PsyArXiv*. <https://doi.org/10.31234/osf.io/va5mk>

Sagehorn, M., Johnsdorf, M., Kisker, J., Sylvester, S., Gruber, T., & Schöne, B. (2023). Real-life relevant face perception is not captured by the N170 but reflected in later potentials: A comparison of 2D and virtual reality stimuli. *Frontiers in Psychology*, 14, 1050892. <https://doi.org/10.3389/fpsyg.2023.1050892>

Salti, M., Monto, S., Charles, L., King, J.-R., Parkkonen, L., & Dehaene, S. (2015). Distinct cortical codes and temporal dynamics for conscious and unconscious percepts. *eLife*, 4, e05652. <https://doi.org/10.7554/eLife.05652>

Saxena, R., Gema, A. P., & Minervini, P. (2025). Lost in Time: Clock and Calendar Understanding Challenges in Multimodal LLMs (Version 2). *arXiv*. <https://doi.org/10.48550/ARXIV.2502.05092>

Schmidt, T. (2000). Visual perception without awareness: Priming responses by color. In *Neural correlates of consciousness: Empirical and conceptual questions* (pp. 157–169). The MIT Press.

Schmidt, T., Lee, X. Y., & Wolkersdorfer, M. P. (2024). Functional dissociations versus post-hoc selection: Moving beyond the Stockart et al. (2025) compromise (Version 3). *arXiv*. <https://doi.org/10.48550/ARXIV.2411.15078>

Schmidt, T., & Vorberg, D. (2006). Criteria for unconscious cognition: Three types of dissociation. *Perception & Psychophysics*, 68(3), Article 3. <https://doi.org/10.3758/BF03193692>

Schwiedrzik, C. M., Singer, W., & Melloni, L. (2011). Subjective and objective learning effects dissociate in space and in time. *Proceedings of the National Academy of Sciences*, 108(11), Article 11. <https://doi.org/10.1073/pnas.1009147108>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3), 417–424. <https://doi.org/10.1017/S0140525X00005756>

Segundo-Ortin, M., & Calvo, P. (2022). Consciousness and cognition in plants. *WIREs Cognitive Science*, 13(2). <https://doi.org/10.1002/wcs.1578>

Sergent, C., Baillet, S., & Dehaene, S. (2005). Timing of the brain events underlying access to consciousness during the attentional blink. *Nature Neuroscience*, 8(10), Article 10. <https://doi.org/10.1038/nn1549>

Sergent, C., & Dehaene, S. (2004). Is Consciousness a Gradual Phenomenon?: Evidence for an All-or-None Bifurcation During the Attentional Blink. *Psychological Science*, 15(11), Article 11. <https://doi.org/10.1111/j.0956-7976.2004.00748.x>

Seth, A. (2021). *Being you: A new science of consciousness*. Dutton.

Seth, A. K. (2025). Conscious artificial intelligence and biological naturalism. *Behavioral and Brain Sciences*, 1–42. <https://doi.org/10.1017/s0140525x25000032>

Seth, A. K., & Bayne, T. (2022). Theories of consciousness. *Nature Reviews Neuroscience*, 1–14. <https://doi.org/10.1038/s41583-022-00587-4>

Seth, A. K., Izhikevich, E., Reeke, G. N., & Edelman, G. M. (2006). Theories and measures of consciousness: An extended framework. *Proceedings of the National Academy of Sciences*, 103(28), 10799–10804. <https://doi.org/10.1073/pnas.0604347103>

Shanks, D. R. (2017). Regressive research: The pitfalls of post hoc data selection in the study of unconscious mental processes. *Psychonomic Bulletin & Review*, 24(3), 752–775. <https://doi.org/10.3758/s13423-016-1170-y>

Shapiro, J. A. (2021). All living cells are cognitive. *Biochemical and Biophysical Research Communications*, 564, 134–149. <https://doi.org/10.1016/j.bbrc.2020.08.120>

Shevlin, H. (2021). Non-human consciousness and the specificity problem: A modest theoretical proposal. *Mind & Language*, 36(2), 297–314. <https://doi.org/10.1111/mila.12338>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Signorelli, C. M., Szczotka, J., & Prentner, R. (2021). Explanatory profiles of models of consciousness—Towards a systematic classification. *Neuroscience of Consciousness*, 2021(2), niab021. <https://doi.org/10.1093/nc/niab021>

Simpson, G. G. (1945). The principles of classification and a classification of mammals. *Bulletin of the AMNH* ; v. 85. <http://hdl.handle.net/2246/1104>

Sims, M. (2024). The Principle of Dynamic Holism: Guiding Methodology for Investigating Cognition in Nonneuronal Organisms. *Philosophy of Science*, 91(2), 430–448. <https://doi.org/10.1017/psa.2023.104>

Singer, P. (2023). *Animal Liberation Now*. Random House.

Singhal, I., & Srinivasan, N. (2024). Just one moment: Unifying theories of consciousness based on a phenomenological “now” and temporal hierarchy. *Psychology of Consciousness: Theory, Research, and Practice*, No Pagination Specified-No Pagination Specified. <https://doi.org/10.1037/cns0000393>

Sligte, I. G., Vandenbroucke, A. R., Scholte, H. S., & Lamme, V. (2010). Detailed Sensory Memory, Sloppy Working Memory. *Frontiers in Psychology*, 1. <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2010.00175>

Spener, M. (2024). *Introspection: First-Person Access in Science and Agency* (1st ed.). Oxford University PressOxford. <https://doi.org/10.1093/oso/9780198867449.001.0001>

Standish, R. K. (2013). Ants Are Not Conscious. *Open Journal of Philosophy*, 03(01), 1–4. <https://doi.org/10.4236/ojpp.2013.31001>

Stockart, F., Schreiber, M., Amerio, P., Carmel, D., Cleeremans, A., Deouell, L., Dienes, Z., Eloseg, P., Gayet, S., Goldstein, A., Halchin, A.-M., Hesselmann, G., Kimchi, R., Lamy, D., Loued-Khenissi, L., Meyen, S., Micher, N., Pitts, M., Salomon, R., ... Mudrik, L. (2025). Studying unconscious processing: Contention and consensus. *PsyArXiv*. https://doi.org/10.31234/osf.io/bkxzh_v3

Storm, J. F., Klink, P. C., Aru, J., Senn, W., Goebel, R., Pigorini, A., Avanzini, P., Vanduffel, W., Roelfsema, P. R., Massimini, M., Larkum, M. E., & Pennartz, C. M. A.

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

(2024). An integrative, multiscale view on neural theories of consciousness. *Neuron*, 112(10), 1531–1552. <https://doi.org/10.1016/j.neuron.2024.02.004>

Sun, H., Zhao, L., Wu, Z., Gao, X., Hu, Y., Zuo, M., Zhang, W., Han, J., Liu, T., & Hu, X. (2024). Brain-like Functional Organization within Large Language Models (No. arXiv:2410.19542). arXiv. <https://doi.org/10.48550/arXiv.2410.19542>

Tal, E. (2020). Measurement in Science. In E. N. Zalta (Ed.), *The Stanford Encyclopedia of Philosophy* (Fall 2020). Metaphysics Research Lab, Stanford University. <https://plato.stanford.edu/archives/fall2020/entries/measurement-science/>

Thompson, D. (1965). Can a machine be conscious?1. *The British Journal for the Philosophy of Science*, 16(61), 33–43. <https://doi.org/10.1093/bjps/XVI.61.33>

Thompson, E. (2015). *Waking, dreaming, being: Self and consciousness in neuroscience, meditation, and philosophy*. Columbia University Press.

Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5(1). <https://doi.org/10.1186/1471-2202-5-42>

Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: From consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), Article 7. <https://doi.org/10.1038/nrn.2016.44>

Trübutschek, D., Marti, S., Ojeda, A., King, J.-R., Mi, Y., Tsodyks, M., & Dehaene, S. (2017). A theory of working memory without consciousness or sustained activity. *eLife*, 6, e23871. <https://doi.org/10.7554/eLife.23871>

Turing, A. M. (1950). Computing machinery and intelligence. *Mind*, LIX(236), 433–460. <https://doi.org/10.1093/mind/LIX.236.433>

Tye, M. (2017). Philosophical Problems of Consciousness. In *The Blackwell Companion to Consciousness* (pp. 17–31). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781119132363.ch2>

Valenzuela-Moguillansky, C., & Vásquez-Rosati, A. (2019). An Analysis Procedure for the Micro-Phenomenological Interview. *Constructivist Foundations*, 14(2), Article 2.

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Van Duijn, M., Keijzer, F., & Franken, D. (2006). Principles of Minimal Cognition: Casting Cognition as Sensorimotor Coordination. *Adaptive Behavior*, 14(2), 157–170. <https://doi.org/10.1177/105971230601400207>

van Gaal, S., Naccache, L., Meuwese, J. D. I., van Loon, A. M., Leighton, A. H., Cohen, L., & Dehaene, S. (2014). Can the meaning of multiple words be integrated unconsciously? *Philosophical Transactions of the Royal Society B: Biological Sciences*, 369(1641), Article 1641. <https://doi.org/10.1098/rstb.2013.0212>

Van Gaal, S., Ridderinkhof, K. R., Fahrenfort, J. J., Scholte, H. S., & Lamme, V. A. F. (2008). Frontal Cortex Mediates Unconsciously Triggered Inhibitory Control. *The Journal of Neuroscience*, 28(32), 8053–8062. <https://doi.org/10.1523/JNEUROSCI.1278-08.2008>

van Gaal, S., Ridderinkhof, K. R., Scholte, H. S., & Lamme, V. A. F. (2010). Unconscious Activation of the Prefrontal No-Go Network. *Journal of Neuroscience*, 30(11), Article 11. <https://doi.org/10.1523/JNEUROSCI.2992-09.2010>

Van Gaal, S., Ridderinkhof, K. R., Scholte, H. S., & Lamme, V. A. F. (2010). Unconscious Activation of the Prefrontal No-Go Network. *The Journal of Neuroscience*, 30(11), 4143–4150. <https://doi.org/10.1523/JNEUROSCI.2992-09.2010>

Van Vugt, B., Dagnino, B., Vartak, D., Safaai, H., Panzeri, S., Dehaene, S., & Roelfsema, P. R. (2018). The threshold for conscious report: Signal loss and response bias in visual and frontal cortex. *Science*, 360(6388), 537–542. <https://doi.org/10.1126/science.aar7186>

Varela, F. J. (1996). Neurophenomenology: A methodological remedy for the hard problem. *Journal of Consciousness Studies*, 3(4), 330–349.

Varela, F. J., Rosch, E., & Thompson, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. The MIT Press. <https://doi.org/10.7551/mitpress/6730.001.0001>

Vorberg, D., Mattler, U., Heinecke, A., Schmidt, T., & Schwarzbach, J. (2004). Invariant Time-Course of Priming with and Without Awareness. In C. Kaernbach, E. Schröger, & H. Müller (Eds.), *Psychophysics Beyond Sensation: Laws and Invariants of Human Cognition*. Psychology Press.

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Wallace, R. (2005). *Consciousness: A Mathematical Treatment of the Global Neuronal Workspace Model*. Springer US. <https://doi.org/10.1007/b106970>

Wang, R., Fu, Y., Chen, L., Chen, Y., Zhou, J., & Chen, H. (2021). Consciousness can overflow report: Novel evidence from attribute amnesia of a single stimulus. *Consciousness and Cognition*, 87, 103052. <https://doi.org/10.1016/j.concog.2020.103052>

Wang, Y., Deng, Y., Wang, G., Li, T., Xiao, H., & Zhang, Y. (2025). The fluency-based semantic network of LLMs differs from humans. *Computers in Human Behavior: Artificial Humans*, 3, 100103. <https://doi.org/10.1016/j.chbah.2024.100103>

Whyte, C. J., & Smith, R. (2021). The predictive global neuronal workspace: A formal active inference model of visual consciousness. *Progress in Neurobiology*, 199, 101918. <https://doi.org/10.1016/j.pneurobio.2020.101918>

Wickens, S. (2022). Review of the evidence of sentience in cephalopod molluscs and decapod crustaceans. *Animal Welfare*, 31(1), 155–156. <https://doi.org/10.1017/S0962728600009866>

Williford, K. (2019). Self-Acquaintance and Three Regress Arguments. *ProtoSociology*, 36, 368–412. <https://doi.org/10.5840/protosociology20193615>

Worrall, J. (1989). Structural Realism: The Best of Both Worlds? *Dialectica*, 43(1–2), 99–124. <https://doi.org/10.1111/j.1746-8361.1989.tb00933.x>

Yang, Y.-H., & Yeh, S.-L. (2018). Unconscious processing of facial expression as revealed by affective priming under continuous flash suppression. *Psychonomic Bulletin & Review*, 25(6), 2215–2223. <https://doi.org/10.3758/s13423-018-1437-6>

Yaron, I., Melloni, L., Pitts, M., & Mudrik, L. (2022). The ConTraSt database for analysing and comparing empirical studies of consciousness theories. *Nature Human Behaviour*, 6(4), 593–604. <https://doi.org/10.1038/s41562-021-01284-5>

Yoshimi, J. (2007). Mathematizing phenomenology. *Phenomenology and the Cognitive Sciences*, 6(3), 271–291. <https://doi.org/10.1007/s11097-007-9052-4>

Section 7: The Outlook of Consciousness Science | Chapter 7.2: Trends in Consciousness Research

Zacks, O., & Jablonka, E. (2023). The evolutionary origins of the Global Neuronal Workspace in vertebrates. *Neuroscience of Consciousness*, 2023(1).
<https://doi.org/10.1093/nc/niad020>

Zheng, Z., Chis-Ciure, R., Waade, P. T., Eiserbeck, A., Aru, J., Andrillon, T., Jarraya, B., Melloni, L., Northoff, G., Rosas, F., & Dwarakanath, A. (2025). Recurrency as a Common Denominator for Consciousness Theories. *PsyArXiv*.
https://doi.org/10.31234/osf.io/wqnzc_v1

Zheng, Z., Trübutschek, D., Huang, S., Cai, Y., & Melloni, L. (2025). What you saw a while ago determines what you see now: Extending awareness priming to implicit behaviors and uncovering its temporal dynamics. *Cognition*, 259, 106104.
<https://doi.org/10.1016/j.cognition.2025.106104>

Zheng, Z.-F., Huang, S.-Y., Lu, S., & Cai, Y.-C. (2023). Interaction between top-down decision-driven congruency effect and bottom-up input-driven congruency effect is correlated with conscious awareness. *Journal of Experimental Psychology: General*.
<https://doi.org/10.1037/xge0001483>